



Microsyntax meets macrosyntax: Russian *neg*-words revisited

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Accepted: 16 February 2024

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Abstract

This paper offers a new analysis of Russian syntactic idioms consisting of stressed general negation *n'e-* fused with a *wh*-word (*k*-word). The elements from this class take infinitival complements and select dative subjects. The clauses with Russian *neg*-words like *mne negde spat* 'I have no space to sleep' and their affirmative counterparts represent the modal existential construction conveying the meaning '*p* is (not) available & *X* can (not) do *q*'. I argue that while the perspective of checking Russian modal existentials on a class of embedded *wh*-infinitives is important, it must be complemented by a comparison of idioms of the *mne negde spat* type with two productive sentence patterns—dative-predicative and dative-infinitive structures. The former are control structures, where dative subjects are matrix clause elements, while the latter have raising properties. Syntactic idioms display mixed properties: on the one hand, they match the overt syntax of dative predicatives, on the other hand, show residual raising effects and license derived non-animate subjects. Like root dative-infinitive structures, syntactic idioms express the meaning of external (alethic) modality, but the same type of modality can be expressed by some dative predicatives. The clauses with *neg*-words originated as embedded dative-infinitive structures, a type marginally acceptable in Modern Russian, while the dative-predicative construction extends its coverage and assimilates *neg*-words. The *neg*-words are derived by the movement of *k*-words into the matrix clause. If a case-marked *k*-word raises to a non-argument position, it loses morphological case and the *neg*-word is reanalyzed as a predicative. If a case-marked *k*-word raises to the subject position, the *neg*-word inherits the case of the *k*-word, which is possible only for dative *k*-words *komu* and *čemu*.

1 *Neg*-words as a problem of Russian grammar

Modern Russian has a class of predicative items consisting of stressed general negation *ne* and a *wh*-element, e.g., the oblique case forms of the interrogative-relative *wh*-pronouns *kto* 'who' and *čto* 'what', cf. *komu*_{DAT}, *kogo*_{ACC-GEN}, *čemu*_{DAT}, *čego*_{ACC-GEN}, etc. or *wh*-adverbials *gde* 'where', *kuda* 'whither', 'where to', *kogda* 'when', *otkuda* 'whence', 'where from', *začem* 'what for', 'why'. Traditional Russian lexicography treats them as words with

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a modal existential meaning. Ožegov's dictionary (1964, p. 395) introduces *nekuda* by the definition "there is no such place where one (could go or put something, etc.)", which is in line with recent semantic research (Kondrashova, 2008; Šimik 2009, 2011; Fortuin, 2014). Following in the footsteps of Rappaport (1986) and Apresjan and Iomdin (1989, 2010), I call the *wh*-elements contained in these items '*k*-words', while the structure into which *k*-words are merged is called '*neg*-words'. These designations do not disclose, whether the stressed *ne* and the *wh*-element form an affix-to-root or clitic-to-root or word-to-word relation, which is a matter of debate. This paper has the following structure. In Sect. 1, I focus on descriptive issues, while in Sect. 2 I discuss several theoretical claims concerning the syntax of *neg*-words and the semantics-to-syntax interface in modal-existential sentences. In Sect. 3, I propose a new model deriving Russian *neg*-words and explore the constraints on their internal structure, i.e., microsyntax in terms of Apresjan and Iomdin (2010). Finally, in Sect. 4, I discuss the predictions made by parametric models of Russian grammar and the status of *neg*-words respective two major sentence patterns—dative-infinitive structures and dative-predicative structures. This macrosyntactic perspective shows how syntactic idioms arise and develop. The general challenge was aptly formulated by Rappaport (1986, p. 81) almost fifty years ago: "...how is this construction <with Russian *neg*-words> to be assimilated with the more familiar patterns of syntactic structure?". I substantiate my answer not by cross-linguistic comparisons of *wh*-constructions but by parametric models of the semantics-to-syntax interface, notably the correlations between types of modality and syntactic structure.

One and the same data often gives rise to different theories. To simplify the navigation through my paper, I first state that I am basing on the description of the Russian verb *byt* in Apresjan (1996) and adopt the distinction of simple negation *ne*¹, which lacks stress and the stressed existential negation *ne*² pace Apresjan and Iomdin (1989, 2010): *ne*² is recognized as the head of the negative Modal Existential Construction (MEC). I also adopt the raising analysis of *k*-words in line with the tradition in Slavic studies starting from Rudin (1986). The novel observations made in this paper focus on three issues: a) Russian *neg*-words take inanimate dative subjects on a regular basis; b) not all uses of Russian predicate *neg*-words pattern with MEC, but all subject *neg*-words do; c) there are grammatical sentences, where the infinitive takes a cased-marked complement already filled by the *k*-word at the start of the derivation. To explain a), I advance the hypothesis that inanimate and non-experiential dative subjects raise from embedded infinitival clauses; b) is explained by the impact of the productive dative-predicative structures (DPS). To explain c), I postulate the deactivation of the case feature with the merger of syntactic heads. I argue that despite the case ending of the *k*-word being visible, there are reasons to treat Russian *neg*-words as caseless expressions approaching the status of DPS predicates, which lack case.

The single arrow ' $\rightarrow$ ' stands for the dependency relation, ' X^0 ' stands for a syntactic head, and the double arrow ' $\Rightarrow$ ' stands for transformation linking the source and the derived structure or for transition from *X* to *Y*.

1.1 Russian *neg*-words and their counterparts

Russian *neg*-words like *nekomu*, *nekogo*, *nečemu*, *nečego*, *negde*, *nekuda*, *nekogda*, *neotkuda*, *nezačem* have three non-trivial features—(i) they select infinitival complements, cf. *nekogo vinit*_{INF} '<there is> no one to blame', *negde spat*_{INF} '<there is> no place to sleep'; (ii) they are not used in non-predicative positions (Paducheva, 2017) and, except for *nekogda*, lack side-uses as standard indefinites, cf. the meanings conveyed by the predicative *nekogda*¹ in (1a–b) vs the meaning of indefinite adverbial quantifier *nekogda*² in (2); (iii)

the general negation *ne* cannot be separated from the *k*-word by any intervening sentence elements,¹ cf. (3a–b) illustrating *negde* ‘there is no such place, where *X* can do *p*’ in present, past and future tenses.

- (1) Russian
 - a. Mne_{DAT.1SG} $\emptyset^{BE.PRES}$ **nekogda**¹_{NEG-W} čitat’_{INF} pis’m_{ACC.PL}.
‘I do not have time to read letters.’
 - b. Mne_{DAT.1SG} $\emptyset^{BE.PRES}$ **nekogda**¹_{NEG-W}.
‘I do not have time’, ‘I am busy.’
- (2) Russian
Nekogda²_{INDEF ONOM.3SG.M} byl_{PST.3SG.M} xorošim_{ADJ.INSTR.SG.M} igrokom_{ADJ.INSTR.SG.M}.
‘He was **once** a good player.’
- (3) Russian
 - a. Mne_{DAT.1SG} $\emptyset^{BE.PRES}$ negde_{NEG-W} spat’_{INF}.
‘I have no place to sleep <now>.’
 - b. Mne_{DAT.1SG} bylo_{PST.3SG.N} negde_{NEG-W} spat’_{INF}.
‘I had no place to sleep.’
 - c. Mne_{DAT.1SG} budet_{FUT.3SG.N} negde_{NEG-W} spat’_{INF}.
‘I will have no place to sleep.’

The tense marker is realized in the past and future by overt copulas² in (3b–c) but by a zero present tense form $\emptyset^{BE.PRES}$ in (1a–b) and (3a). This distribution is characteristic of tense forms of *byt’* in general (Apresjan, 1996, pp. 518, 528). The zero form $\emptyset^{BE.PRES}$ is not specified in Modern Russian as copula vs content verb, i.e., existential, locative, or possessive predicate, cf. the possessive structure *U menja*_{GEN.PREP} $\emptyset^{BE.PRES}$ *kvartira*_{NOM} *v gorode N*. ‘I have an apartment in the city of N.’ alternating with the variant with overt affirmative form *est’*_{BE.PRES} (Dymarskij, 2018; Letuchiy, 2018a; Zimmerling, 2021a, p. 10). The overt form *ecm*_{BE.PRES} does not occur with *neg*-words, cf. (4a). This might be due to a semantic anomaly, since the negative existential operator *ne* ‘there is not any *x*’ licensing the *k*-word and the affirmative existential operator *est’* ‘there exists some *x*’ are not expected to combine. However, the negative counterpart of *est’*, the predicative negative marker *net* ‘no’, ‘not’, ‘there is not any *x*’ is generally blocked in Standard Russian in the context of *k*-words as well, cf. (4b).

¹Except for the prepositions splitting the oblique forms of *neko*, *nečto*, cf. **uprep nekogo*_{GEN} *otprašivat’ šja*_{INF} ‘to ask anybody for leave’ $\Rightarrow$ *ne*_{NEG} *uprep kogo*_{GEN} *otprašivat’ šja* ‘<*X*> has no one to ask for leave’. Split oblique forms of indefinites with *koe-* (*u*, *koe-kakoj* $\Rightarrow$ *koe u kakogo*) and non-predicative negatives with *ni-* (*u*, *nikto* $\Rightarrow$ *ni u kogo*) are a well-known feature of Modern East Slavic. The splitting of *koe*-indefinites and *ni*-negatives can be interpreted as endoclitisis, since *koe-* and *ni-* lack word properties (Arkadijev, 2016). For Russian *neg*-words, an endoclitisis analysis depends on the assumption that *ne-* is a standard prefix.

²The copular forms of *byt’* occasionally alternate with semi-content verbs *stat’*, *okazat’ šja*, *delat’ šja* marking the change of state (Apresjan & Iomdin, 2010, pp. 92–94). Apresjan & Iomdin also mention *najtis’* ‘show up’, but it is licensed only with distant negation, cf. (i), not with *neg*-words, cf. (ii).

(i) **Ne**_{NEG} našlos’_{AUX.PST.SG.N}, [**kogo**_{WH-ACC} poslat’_{INF}].
‘There was no one to send.’

(ii) ***Nekogo**_{NEG-W} našlos’_{AUX.PST.SG.N} poslat’_{INF}.

Fortuin (2014, pp. 32, 52) provides several examples with *najdetsja* intervening between *ne* and the *k*-word.

- (4) Russian
- *Est'**_{BE.PRES} **negde**_{NEG-W} spat'_{INF}.
 - *Net**_{NEG}, gde_{K-W} spat'_{INF}.
Int. 'I have no place to sleep.'

In the context of an interrogative operator expressed by the yes-no particle *li*, the combination of *k*-word with *net* is marginally acceptable (Apresjan and Iomdin 1989, 2010, p. 72), though not with all predicates.

- (5) Russian
- Net_{NEG} **li**_Q u tebj_{GEN.PREP}, [čego_{K-W·ACC-GEN} vypit'_{INF}]?'
'Do you have anything to drink?'
 - Net_{NEG} **li**_Q u tebj_{GEN.PREP}, [čem_{K-W.INSTR} zapisat'_{INF} nomer_{ACC}]?'
'Do you have anything to write down the phone number?'
 - ?Net_{NEG} **li**_Q u tebj_{GEN.PREP}, [gde_{K-W} perenočevat'_{INF}]?'
'Do you have any place to spend the night?'
 - ??Net_{NEG} **li**_Q u tebj_{GEN.PREP}, gde_{K-W} pospat'_{INF}?
Int. 'Do you have any place to sleep?'

Russian *k*-words express the diagnostic meaning 'there is some *x* / there is not any *x* so that it is possible to make *p*' only if there is a higher lying operator, e.g., the general negation *ne* '∼ *p*' in (3) or the alternative particle *li* '*p* or ∼ *p*'³ in (5), but the derivation of *neg*-words only takes place in negative clauses, where *ne* and the *k*-word make up a contact sequence and end up in the same clause. In all other cases, the operator is realized in the matrix clause, while the *k*-word sits in the embedded clause. This generalization also holds for the affirmative sentences, where the operator *est'*_{BE.PRES} 'there is' does not make up a unit with the *k*-word, cf. (6) and (7). The puzzling thing is that the clauses with *neg*-words require the zero copula ∅_{BE.PRES} in the present indicative (Švedova, 1982, p. 346), while their affirmative correlates are non-copular as shown by the overt content verb form *est'* in the present indicative.⁴

- (6) Russian
Mne_{DAT.1SG} **est'**_{BE.PRES}, gde_{K-W} spat'_{INF}.
'I have a place where to sleep.'
- (7) Russian
Mne_{DAT.1SG} **bylo**_{BE.PST.SG.N}, **est'**_{BE.PRES} i **budet**_{BE.FUT.3SG}, čto_{K-W} čitat'_{INF}.
'I had, have, and will have what to read.'

1.2 The modal existential meaning and Slavic MEC sentences

There are reasons to assume that syntactic contrasts notwithstanding, negative and non-negative existential sentences have similar argument structures in the same language if they

³Colloquial Russian occasionally licenses dropping *li* in the context of *net*, e.g., *Net # u tebj_{GEN} čego vypit'*?. Such sentences express the meaning '*p* or ∼ *p*' with an additional flavor, ca. 'You have something to drink, right?'.

⁴The non-present forms *byt'* are two-way ambiguous between copular vs content-verb uses. The overt present tense form *est'*_{BE.PRES} has pragmatically marked copular uses in bi-nominative sentences like *Vasja*_{NOM.SG} i *est'*_{BE.PRES} *naš načal'nik*_{NOM.SG} 'Basil is indeed our boss', *Ėkzistencializm*_{NOM.SG} *est'*_{BE.PRES} *gumanizm*_{NOM.SG} 'Existentialism is <the same thing as> humanism', but such patterns do not occur in Russian MEC.

share such components as *k*-words, infinitives, and operator positions. The general predicate type behind Russian negative clauses with *neg*-words like (1), (3), alternative *wh*-questions like (5), and affirmative clauses like (6), (7) is *modal existential*. Ožegov's definition of the Russian *neg*-word *nekuda* quoted above in Sect. 1.1. can be generalized for all languages, where the corresponding predicate type is diagnosed, cf. (i).

- (i) There is <a property or event> *p* so that *X* can do *q*.

The term 'modal existential' coined by Grosu (2004) is adopted by Šimík (2011, pp. 66, 106), who proves that modal-existential *wh*-construction (hence—MEC⁵) is similar but not identical to free and headed relatives and indirect *wh*-questions. He also provides paraphrases of (i) proposed in the preceding literature (Šimík, 2011, p. 99). Zimmerling (2022) offers a tag 'hybrid modal existential' for Russian and Bulgarian MEC. The modal existential meaning can be explained differently. Šimík considers it an elementary type of modality and assumes that existential statements are already modal, while other authors decompose it to the modal component coming from the complement infinitival clause and the existential component coming from the matrix clause (Ebeling, 1984). Fortuin (2014, pp. 55–57) argues with some provisos that the meaning of Russian MEC is a composition of three meanings expressed by the infinitive, existential verb and dative subject.⁶

The grammar of Slavic MEC is relatively uniform from the perspective of the semantics-to-syntax interface. Mrázek (1972, pp. 98–101) classifies the attested syntactic types based on three criteria: 1) the availability of independent infinitival clauses including deliberative *wh*-questions: these sentences remain productive in East Slavic but are partially eliminated in other parts of the Slavic area, cf. Russian *Komu_{WH-DAT} paxat'_{INF}*? 'Who will plow?' or 'Whose turn is it to plow?' vs Czech **komu_{WH-DAT} orat_{INF}*?; 2) the use of BE (*esse*) vs HAVE (*habere*) as an existential operator; 3) symmetric vs asymmetric syntax in affirmative and negative MECs. Modern Russian has a peripheral position in Mrázek's scheme: it is a consistent BE-language with root infinitival constructions and asymmetric syntax in negative vs affirmative MECs. Other East Slavic languages along with Polish use HAVE-operators in negative MECs, cf. Belorussian *njama_{HAVE.NEG.3SG} čago_{WH.ACC-GEN} rabic'_{INF}* 'There is nothing to do' (Mrázek, 1972, p. 101). The most consistent HAVE-languages with symmetric MEC syntax are Bulgarian and Macedonian, which replaced *wh*-infinitives with finite *da*-clauses. In the Bulgarian examples (8a–b) the *k*-word *kāde* 'where' stands to the left of the complementizer particle *da*, which indicates that *kāde* is raised (Penchev 1984, 1993, 1998).

(8) Bulgarian

- a. Ima_{EXIST.3SG} *kāde_{WH}* [_{MEC} *da spja_{3SG} kāde*].
'There is a place, where one can sleep.'
- b. Njama_{EXIST.NEG.3SG} *kāde_{WH}* [_{MEC} *da spja_{3SG} kāde*].
'There is no place, where one can sleep.'

Most Slavic languages have internal variation in MEC. Russian is not an exception, since negative MECs in the past and future tense can be realized without merging *k*-words with *ne*, so structures with *neg*-words like those in (3b–c) arguably have variants (9b–c)

⁵The plural form 'MECs' stands for 'MEC sentences'.

⁶The contrast between Šimík's and Fortuin's accounts is not radical: neither of them claims that the inherent lexical meaning of the existential verb remains unmodified in MEC. The former stipulates that BE and HAVE only bring about modal meanings in a group of contexts including MEC but not in other cases, e.g., a sentence of the type 'X has Y' is not modal. The latter argues that the 'modality of EC is an interpretation, which is contextually determined, rather than a meaning expressed by a form' (Fortuin, 2014, p. 55).

(9) Russian

- a. Mne_{DAT.1SG} **ne**_{NEG} bylo_{PST.3SG.N} **gdewh** spat'_{INF}.
'I had no place to sleep.'
- b. Mne_{DAT.1SG} **ne**_{NEG} budet_{FUT.3SG.N} **gdewh** spat'_{INF}.
'I will have no place to sleep.'

The common wisdom about examples like (9a–b) is that they are less idiomatic and much less frequent compared to sentences with *neg*-words (Paducheva, 2017). At the same time, they have a prominent place in linguistic papers, cf. Rappaport (1986), Apresjan and Iomdin (1989), Avgustinova (2003), Fortuin (2014, p. 32), since they are relevant for proving two points: 1) the derivation of Russian *neg*-words is not an obligatory process; 2) there can be a rule combining different word orders with negation. Apresjan and Iomdin (2010, p. 95) supply (3b) and (9a) with the same translation 'there is no such place so that I could sleep in that place' and claim that '...if there is any difference between the sentences *mne negde*_{NEG-W} bylo_{PST.3SG.N} spat'_{INF} and *mne ne*_{NEG} bylo_{PST.3SG.N}, *gdewh* spat'_{INF}, it solely pertains to communicative factors'. However, many sentences with *neg*-words lack natural variants with distant negation. E.g., sentences with copulas *okazat'sja*, *najtis'*, *stat'* and distant negation are well-formed without overt dative subjects in (10a) but weird with them, cf. (10b), while sentences with *neg*-words are acceptable with two copulas from this set, cf. (10c). Similar restrictions hold for clauses without dative subjects, cf. (10d).

(10) Russian

- a. Vdrug **ne**_{NEG} **okazalos'**_{PST.AUX.SG.N} / **našlos'**_{PST.AUX.SG.N} / **stalo**_{PST.AUX.SG.N},
[MEC čem_{WH} platit'_{INF}]. (after Avgustinova, 2003 and Fortuin, 2014, p. 53)⁷
'It turned out to be nothing to pay with.'
- b. */?? Vdrug **nam**_{DAT.1PL} **ne**_{NEG} **okazalos'**_{PST.AUX.SG.N} / **našlos'**_{PST.AUX.SG.N} /
stalo_{PST.AUX.SG.N}, [MEC čem_{WH} platit'_{INF}].
Int: 'It turned out that we have nothing to pay with.'
- c. Vdrug **nam**_{DAT.1PL} **stalo**_{PST.AUX.SG.N} / ?**okazalos'**_{PST.AUX.SG.N} /
***našlos'**_{PST.AUX.SG.N} [MEC **nečem**_{NEG-W} platit'_{INF}].
'Suddenly we had nothing to pay with.'
- d. Vdrug **stalo**_{PST.AUX.SG.N} / ??**okazalos'**_{PST.AUX.SG.N} / ***našlos'**_{PST.AUX.SG.N}
[MEC **nečem**_{NEG-W} platit'_{INF}].
'Suddenly one had nothing to pay with.'

Such discrepancies indicate that the morphosyntax of *neg*-words, i.e., surface position of the negation, *k*-word and spell-out of overt dative subjects has an impact on semantic interpretation. (10a) tells about the lack of property *p* (money), while (10c–d) tell that some referential or weak referential (underspecified) animate subject *X* was short of money in some spatiotemporal situation *p*.

1.3 Neg-words and predicatives

Russian *neg*-words are usually called predicative negative pronouns projecting the sentence pattern PRON_{NEG}—INF (Švedova, 1982, p. 346; Paducheva, 2017). Some authors, cf. Veyrenc (1979, p. 73) however identify them with impersonal predicatives. This tag is motivated by two features classifying them with a large class of non-verbal elements selecting dative-predicative structures (hence—*DPS*) like *mne*_{DAT.1SG} *nel'zja*_{PRED} / *nado*_{PRED} /

⁷Here and in the following the notation is mine. I use agnostic tags InfP and NegWP throughout this paper.

*žal'*_{PRED} / *pora*_{PRED} *uezžat'*_{INF} 'I cannot / have to leave / I am sorry to leave / It is time for me to leave'—a) lack of agreement—in the past tense, the copula with DPS items and *neg*-words assumes the default agreement form (3SG.N) in the absence of a lexical controller: <*mne*_{DAT-1SG}> *bylo*_{AUX.PST.SG.N} *nel'zja*_{PRED} / *nekuda*_{NEG-W} *uezžat'*; b) the animacy condition. The affinity of *neg*-words with modal or experiential predicatives like *nel'zja* 'cannot', *pora* 'it is time', *žal'* 'it is a pity', *oxota* 'desire' taking animate dative subjects was evident already to Alexander Peškovskij, who in 1928 called them 'amorphous words in the position of an impersonal predicative element' (1928, pp. 416–418). Peškovskij points out that DPS predicatives of nominal, cf. *žal'*, *oxota*, *pora*, pronominal, cf. *nekuda*, *neotkuda*, *negde*, *nezačem* or adjectival origin, cf. *nužno*, *možno*, *dolžno* lose the connection with declension paradigms.⁸ A more radical approach has been developed by Peškovskij's contemporary Lev Ščerba (1928), who claimed that Russian predicatives, especially those licensing DPS sentences tend to make up a new part of speech labeled 'category-of-state'. Although Ščerba himself did not mention *neg*-words, his follower Alexander Isačenko (1965, p. 288) added them to the hypothetical category-of-state class. The ontology of the DPS construction can be built without recourse to Ščerba's hypothesis. Zimmerling (2018a) subcategorized Russian DPS predicatives into 15 classes and placed all *neg*-words with modal predicatives (CLASS 2 elements). The class of DPS predicatives includes some idioms like *X-u bylo vse ravno*_{PRED} 'X did not care', 'X was indifferent' (**X-u bylo ravno*), *X-u stalo ne po sebe*_{PRED} 'X felt uncomfortable' (**X-u stalo po sebe*) that make up quasi-synonymic relations with DPS items realized by single word forms, cf. *X-u bylo bezrazlično*_{PRED} 'X was indifferent', *X-u stalo durno*_{PRED} 'X felt bad'. If *neg*-words like *nekogo*, *nekuda* are idiomatic DPS items with inbuilt lexical negation, cf. *ne po sebe* (**po sebe*), *nel'zja* (**l'zja*),⁹ the infinitive must be analyzed as their complement: *nekogo*⁰ → *vinit'*_{INF}, *nekuda*⁰ → *spešit'*_{INF}, *ne o čem*⁰ → *žalet'*_{INF}, *nekem*⁰ → *gordit'sja*_{INF}. However, the infinitive can be the host, not the complement, since the case form of the *k*-word is predicted by the government pattern of the corresponding verb, e.g., *vinit'* assigns ACC, *žalet'* assigns LOC and *gordit'sja* assigns INSTR: *vinit'*⁰ → *kogo*_{ACC}, *žalet'*⁰ → *o čem*_{LOC}, *gordit'sja*⁰ → *čem*_{INSTR}. That makes *neg*-words a 'persistent' problem of Russian grammar as the title of Rappaport (1986) suggests, since their argument structure does not correspond to their overt syntax.

2 State of research

There is a large body of theory-oriented research discussing Russian *neg*-words. It can be broadly divided into three groups: 1) syntax and semantics of negative and non-negative MEC sentences in a class of languages including those from the Slavic group, cf. Zubatý (1922), Mrázek (1972), Izvorski (1998), Pancheva-Izvorski (2000), Grosu (2004), Šimik (2009, 2011), also Kalėdaitė (2008) for Lithuanian and Khatatneh (2020) for Modern Standard Arabic; 2) specific build-up of *neg*-words and Russian negative MECs, cf. Holthusen (1953), Garde (1976), Rappaport (1986), Apresjan and Iomdin (1989, 2010), Babby (2000), Avgustinova (2003), Kondrashova (2008), Kondrashova and Šimik (2013), Fortuin (2014); 3) general conditions licensing infinitives and predicatives in Russian, cf. Moore and Perlmutter (1999, 2000), Perlmutter and Moore (2002), Fortuin (2005, 2007, 2014), Fleischer (2006), Zimmerling (2009, 2018a, 2020a), Madariaga (2011), Israeli (2013, 2014, 2016),

⁸The last part of Peškovskij's point does not apply to adverbial *neg*-words like *negde*.

⁹A similar analysis has been outlined by Paducheva (2017), who claims that structures with lowered, i.e., lexically bound negation lack semantic interpretation.

Gerasimova (2015), Mitrenina (2017), Letuchiy (2018b), Kustova (2021), for a survey of negation positions in Slavic clauses cf. Kosta (2009). That these research fields are specialized is confirmed by the fact that two important works—Apresjan and Iomdin (2010, pp. 62–81) and Šimík (2011, pp. 208–308), which appeared around the same time,¹⁰ discuss earlier publications¹¹ but highlight different details. A criticism of several Russian examples or their translations circulating in the literature can be found in Israeli (2013, pp. 201, 209, 212; 2016, pp. 307–308), Apresjan and Iomdin (2010, p. 98) and Šimík (2011, pp. 141, 199).

2.1 General negation and *k*-words as syntactic elements

There is a long line of research, where *k*-words and the general negation *ne* are treated as syntactic elements. This approach has two motivations—1) the need to explain the asymmetric syntax of Russian negative and affirmative MECs; 2) the need to account for cross-linguistic variation in Slavic negative MECs. According to Mrázek (1972, pp. 97–98) and Šimík (2011, p. 143), the great Czech linguist Josef Zubatý (1922) was the first scholar who argued that Slavic MECs are bi-clausal structures based on deliberative questions like *Čto čítat?* ‘*Kuda pojti?*’ Šimík (2011, p. 289) adds that despite Zubatý treating MECs as embedded *wh*-questions, he noticed some deviations: while Czech blocks clitic climbing out of embedded *wh*-questions, cf. (11b), Czech MECs license it, cf. the position of the reflexive *=se* in (11a).

(11) Czech (Zubatý, 1922, p. 66, quoted upon Šimík, 2011, p. 5)

- a. Mám_{1SG} = *se*_{REFL} _{[MEC nač_{PREP-WH} tešit *t_i*].}
- ‘I have something to look forward to.’
- b. *Vím_{1SG} = *se*_{REFL} _{[Q kam posadit *t_i*].}
- Int. ‘I know where to sit down.’

This observation made by Zubatý hundred years ago supports the idea that indirect questions are islands blocking extractions, while clitic climbing only takes place, if the complement clause is restructured, i.e., loses some properties of an autonomous clause (Stjepanović, 2004). That entails that Slavic MECs as structures with clitic climbing favor a restructuring analysis. A similar approach has been applied to Russian, a language lacking standard object and reflexive clitics and climbing effects by Kondrashova (2008) and Kondrashova and Šimík (2013). They analyze the *wh*-complements of Russian *neg*-words as VP-infinitives, i.e., reduced structures lacking the properties of full-developed complement infinitival clauses (= CP-infinitives). A key fact in favor of this analysis is that Slavic MECs generally allow extraction of non-clitic elements, which was first demonstrated for Bulgarian by Rudin (1986). The same holds for Russian, cf. Rappaport (1986, p. 13). I provide a pair of examples with the extraction of reciprocal pronouns from MECs containing a *neg*-word.

(12) Russian

- a. <Oni vragi>. *Drug drugu*_{REC.DAT i} _{[MEC im_{DAT.3PL} nečego_{NEG-w} bylo_{AUX.PST.SG.N} skazat’_{INF} ____i.].}
- ‘<They are enemies.> They had nothing to say to each other.’

¹⁰ The chapter Apresjan and Iomdin (2010) is a revised and expanded version of Apresjan and Iomdin (1989).

¹¹ Šimík’s thesis (2011) seems to be the only work in the field that includes an annotated bibliography.

- b. <Oni druž'ja>. *Drug ot druga*_{REC.GEN.PREP i} [_{MEC im}_{DAT.3PL} *nečego*_{NEG-W} *bylo*_{AUX.PST.SG.N} *skryvat'*_{INF} _____i].
'<They are friends.> They had nothing to conceal from each other.'

According to Apresjan and Iomdin (2010, p. 70), Johannes Holthusen (1953, p. 158) was the first linguist who analyzed Russian *neg*-words as syntactic structures and treated *ne-* as a negative existential verb. A more detailed version of this approach comes from Garde (1976). Both Holthusen and Garde assume that *k*-words are heads in the pair (*k*-word⁰, infinitive) and postulate the hierarchy (ii) for Russian negative MECs.

- (ii) $ne^0 \rightarrow k\text{-word}^0 \rightarrow \text{infinitive}$. (after Holthusen, 1953 and Garde, 1976)

Garde claims that the subject dative argument in Russian MEC directly depends on the existential verb. The validity of this analysis is denied by Apresjan and Iomdin (2010, p. 81), who add a Tense node¹² above the infinitive and treat DAT as the lowest complement: ...Tense⁰ → Infinitive⁰ → DAT. However, the dative case can be assigned both by the predicatives or verbs selecting dative arguments and by infinitives. The subjects of DPS predicatives are assigned lexically and are matrix clause elements (Zimmerling, 2009), while the subjects of the dative-infinitive-structures (hence—*DIS*) like *mne*_{DAT.ISG} *ešče sidet'*_{INF} *nočju* 'I have to sit up for some time at night' are not assigned lexically and raise to a higher clause, cf. Babby (2000), Perlmutter and Moore (2002), Fleisher (2006), Mitrenina (2017), Zimmerling (2021a, p. 279). Hence follows that if Russian *neg*-words have DPS syntax, an analysis à la Garde is an option, but if they have DIS syntax, Apresjan & Iomdin's approach is strongly preferable.

Rappaport (1986) calls Russian *neg*-words 'syntactic idioms', i.e., elements that have internal syntactic structure but are stored in the lexicon. He argues that MECs are headed by a two-place existential predicate BE that takes a dative NP as its external argument (= subject in the traditional notation) and a *k*-word as its internal argument (= object in the traditional notation). In the pair (infinitive⁰, *k*-word) the infinitive is the syntactic head. However, the semantic structure is different: the infinitive is the argument of the *k*-word. That makes MECs with *neg*-words an idiosyncratic construction. *K*-words raise from embedded clauses, which makes it possible to sum up Rappaport's analysis by a simplified scheme (iii):

- (iii) $ne^0 \rightarrow k\text{-word}^0 \rightarrow \text{infinitive}^0 \rightarrow \text{k-word}$. (after Rappaport, 1986).

Rappaport does not derive Russian *neg*-words in syntax and assumes that *ne-* and *k*-word form a unit before sentence derivation takes place. This view is adopted by Avgustinova (2003), who uses the HPSG formalism but is generally rejected in minimalist syntax. The assumption that all Slavic MECs are derived by *wh*-movement, cf. Izvorski (1998), Pancheva-Izvorski (2000)¹³ is shared by Babby (2000), who argues that Russian *neg*-words are formed post-syntactically after *k*-words raise into the matrix clause and merge with *ne-*. He also claims that dative subjects of *neg*-words come from embedded clauses and are raised. Babby's analysis is rejected by Fleisher (2006), who argues that dative subjects are matrix clause elements, which control PRO in the embedded infinitival clause. Fleisher's analysis revives the much older proposal by Garde (1976), albeit in a modern disguise.

Apresjan and Iomdin (1989, 2010) apply the dependency formalism of the Meaning-Text Theory and outline a shared structure for Russian negative and affirmative MECs. It includes

¹² The label 'Tense' is not used in Apresjan and Iomdin (2010) but is a generalization over specific tags used by them in the annotation of dependency trees on pp. 78–80.

¹³ Generative scholars usually treat the orders like #...*wh* ...*Inf* as derived even if the *wh*-element is not fronted to the clausal left periphery.

five positions and is headed by the existential predicate. The latter must be overt and is realized either as a non-zero form of the content verb BE (in affirmative clauses and clauses with the distant negation of the type *ne bylo, gde spat'*) or by the merged negation *ne-*. It is supplied by the superscript ne^2 to keep it apart from the negative particle ne^1 and is interpreted as a negative existential verb. Unlike Rappaport, they treat BE as a one-place predicate, its only complement being the node Tense filled either by an overt copula or by the zero form $\emptyset^{BE.PRES}$ depending on the tense value and the copula. Tense dominates over two sister nodes—Infinitive and [Variable]: the latter stands for different *k*-words, e.g., *gde, kogda, kogo* or for semantic arguments they introduce, e.g., [Place], [Time], [Theme], etc. If the dative argument is present, it is treated as an infinitival complement. This scheme is rendered below in a modified notation (iv).

- (iv) $EXIST/NEG.EXIST^0$ (Content verb BE $\vee ne^2$) $\rightarrow$ $TENSE^0$ (Overt copula $\vee \emptyset^{BE.PRES}$) $\rightarrow$ Infinitive, VARIABLE; infinitive $^0 \rightarrow$ DAT. (adapted from Apresjan and Iomdin 1989, 2010)

Apresjan & Iomdin dub the *neg*-words ‘syntactic agglomerates’ ($\approx$ Rappaport’s ‘syntactic idioms’) but do not specify, whether they are derived post-syntactically or pre-syntactically.¹⁴ Fortuin (2014, p. 52) criticizes the identification of ne^2 with an existential verb and calls their analysis with two verbal centers $ne^2 + bylo/budet$ ‘rather inelegant’. However, it captures two observations that hold for all known languages with MECs: 1) MECs are always headed by an *overt* (non-zero) operator; 2) the hierarchy of EXIST and TENSE nodes corresponds to the structure of a polypredicate complex. Although Apresjan & Iomdin do not stick to the notion of a clause, their model is in line with bi-clausal accounts by Rappaport (1986), Babby (2000) and subsequent authors. A large part of Apresjan & Iomdin’s analysis is adopted by Avgustinova (2003) and Kondrashova and Šimik (2013),¹⁵ while Paducheva (2017) outlines a version that calls back the ideas of Garde (1976). Following Garde, she assumes that the full form of the negative existential predicate is *net*, not ne^2 and the underlying structure of the sentence *Detjam negde spat'* is **Detjam net, gde spat'*. The existential operator in negative MECs is silent, it is a two-place predicate which dominates over the negation and the node VARIABLE. This tag is not used by Paducheva, but she specifies that the corresponding node stands for the type of semantic argument mapped to the *k*-word, e.g., for *gde* ‘where’ it will be [Place]. The node VARIABLE takes an infinitive as its complement (VARIABLE $^0 \rightarrow$ infinitive). The infinitive has two complements: the dative subject and the *k*-word mapped to VARIABLE.

- (v) $\langle silent \rangle OPERATOR^0 \rightarrow NEGATION, VARIABLE; VARIABLE^0 \rightarrow Infinitive^0 \rightarrow DAT, k\text{-word}$ (adapted from Paducheva, 2017)

Paducheva mentions a similarity between Russian MECs and free relatives of the type *Ja s'em, [FR čto_{WH} ty prigotoviš]* ‘I’ll eat [FR what you prepare]’. Although she does not identify them directly, she argues that such interpretation is desirable, since a special position for VARIABLE is no longer needed if sentences like *detjam ne bylo gde spat'* are free relatives. Paducheva supports her claim by examples with distant position of the *k*-words and negation

¹⁴ Avgustinova (2003) and Kondrashova and Šimik (2013, Sect. 4.2.) ascribe to Apresjan & Iomdin the idea that *neg*-words are assembled before syntactic derivation, but that is merely a framework-internal issue. The Meaning-Text Theory used by Apresjan & Iomdin lacks morphological rules that are applied post-syntactically or parallel with syntactic operations.

¹⁵ Apresjan & Iomdin’s analysis of *neg*-words is implemented in the deep annotation of SynTagRus corpus, which is part of the Russian National Corpus (RNC) and the Universal Dependencies project.

like (9a–b). The identification of MECs with free relatives is based on shaky ground: Rudin (1986) denies that MECs and free relatives make up the same class in Bulgarian and Šimík (2011, pp. 90–91) arrives at the same conclusion for a sample of 24 languages.

2.2 Semantic interpretation of MEC sentences and Russian *neg*-words

There are two main issues in semantic analysis of MEC—existential quantification and realization of the modal component. MEC is a natural language quantifier, but its type can be interpreted differently. Šimík (2009, 2011, p. 308) classifies the proposals into three groups, with the proviso that the argumentation of some authors is at odds with their formalism: 1) MECs are treated as *generalized quantifiers*, i.e., expressions of the type $\langle \langle e, t \rangle t \rangle$ that are subject to quantifier raising (Rappaport, 1986; Grosu, 2004). Under this analysis, MECs resemble quantified DPs used in argument positions; 2) MECs are treated as *properties*, i.e., expressions of the type $\langle e, t \rangle$ by virtue of the left-peripheral, i.e., raised *wh*-operator, their quantificational force comes from the matrix verb (Izvorski, 1998; Pancheva-Izvorski, 2000). Under this analysis, MECs resemble predicative NPs or relative clauses; 3) MECs are treated as *open propositions*, i.e. expressions of the type $\langle t \rangle$ or $\langle s, t \rangle$ that are open by virtue of the presence of an unbound variable realized by the *wh*-word. The apparent quantificational force comes from the matrix verb (Šimík, 2009; Kondrashova & Šimík, 2013). Under this analysis, MECs are similar to embedded questions. Kondrashova and Šimík (2013) adjust this classification to Russian data. They do not mention the hypothesis that MECs express properties but discuss a fourth approach called Unselective Binding Analysis, whereby the verb BE is treated as an unselective existential quantifier and the *wh*-words are free variables. The motivation comes from the fact that Slavic languages normally license multiple *wh*-elements in MECs (Rudin, 1986), which also holds for Russian affirmative (13a) and negative (13b) MECs.

(13) Russian (Kondrashova & Šimík, 2013)

- a. Maše bylo_{AUX.PST.3SG.N} s kem_{WH-PREP.INSTR} o čem_{WH-PREP.INSTR} pogovorit'_{INF}.
'There was someone for Masha to talk with about something.'
- b. Maše ne bylo_{AUX.PST.3SG.N} s kem_{WH-PREP.INSTR} o čem_{WH-PREP.INSTR} pogovorit'_{INF}.
'There was someone for Masha to talk with about something.'

The novel observation made by Kondrashova & Šimík is that multiple *wh*-elements are blocked in Russian, if MEC contains a *neg*-word, cf. (14). Therefore *neg*-words cannot be described in terms of unselective binding.

(14) Russian (Kondrashova & Šimík, 2013)

- *Maše ne s kem_{NEG-W} bylo_{AUX.PST.3SG.N} o čem_{WH-PREP.INSTR} pogovorit'_{INF}.
Int. 'There was no one for Masha to talk with about something'

Generalized quantifiers in the wake of Barwise and Cooper (1981) are selective but not domain-specific. A Generalized Quantifier analysis correctly predicts the ill-formedness of (14), since the variable represented by the second *wh*-word *o čem* remains unbound. However, it fails to capture two other constraints—the ill-formedness of the combination *neg-word* + *est*', cf. (15a) and the earlier provided test (4a) and the impossibility of multiple *neg*-words in a single-clause, cf. (15b). Adding conjunction *i* 'and' in the sequence Neg-W1

Neg-W2 would save (15b), but that only confirms that Russian has a ‘one *neg*-word per clause constraint’.¹⁶

(15) Russian (Kondrashova & Šimík, 2013)

- a. *Maše_{DAT} **ne s kem**_{NEG-W} **est**_{BE.PRES} **o čem**_{NEG-W} *pogovorit*_{INF}.
Int. ‘There was no one for Masha to talk with about.’
- b. *Maše_{DAT} **ne s kem**_{NEG-W} **ne o čem**_{NEG-W} *pogovorit*_{INF}.
Int. ‘There is no one whom Masha could speak about nothing.’

Thus, the Unselective Binding hypothesis only predicts the syntax of Russian negative MEC with distant negation and the syntax of affirmative MEC, while the Generalized Quantification hypothesis predicts the syntax of *neg*-words but does not capture their relation to presumably synonymic MEC with distant negation. This result is not desirable, and Kondrashova & Šimík outline a calculus with propositional quantifiers that are both selective and domain-specific. Their model based on Hamblin (1973), Kratzer and Shimoyama (2002) and Yanovich (2005) approaches the complexity of natural language. Still, they admit that their analysis leaves several issues, e.g., why *ne*- and *net* are not equivalent, if both of them mean ‘ $\sim (\exists (x))$ ’ or why *net* cannot be replaced by the combination **ne est*’, cf. the impossibility of **U menja ne est* ‘brata, int. ‘I do not have a brother’.¹⁷

The variety of approaches to defining the modal component of MEC (Fortuin 2005, 2007) is less pronounced. Šimík (2011, pp. 55–56) based on a sample of languages and Zimmerling (2022) for Russian and Bulgarian treat the taxonomic type of MEC similarly but stick to different terms: root modality that does not express deontic, epistemic, bouletic meanings and ability (Šimík), alethic/external modality (Zimmerling). Both authors state that its modal force is weak. That means that they only express *external possibility* ($\Diamond p$) or *external impossibility* ($\sim \Diamond p$): there is no evidence that MEC can have strong modal force and express external necessity ($\Box p$). The meaning of external possibility is not unique for Russian MEC, see (16a), it can be expressed by DIS sentences, cf. (16b) and by DPS predicatives like *nevozmožno* ‘impossible’, *nado* ‘need’, cf. Zimmerling (2009) and Mitrenina (2017, p. 68).

(16) Russian

- a. Mne_{DAT.1SG} **negde**_{NEG-W} *vyexat*_{INF}. [MEC]
‘There is no place for me to drive from here.’
- b. Mne_{DAT.1SG} **ne**_{NEG} *vyexat*_{INF}. [DIS]
‘I cannot drive from here.’
- c. Mne_{DAT.1SG} **nevozmožno**_{PRED} *vyexat*_{INF}. [DPS]
‘It is not possible for me to drive from here.’

The sentences (16a–b) share the same sort of modality ‘External Force prevents *X* from doing *p* in spatiotemporal situation *S*’. That does not entail that MEC, DIS and DPS are synonymic. Root DIS sentences can also signal *external necessity* ($\Box p$), cf. Fortuin (2005): as Mitrenina (2017) points out, this option is reserved for affirmative root DIS sentences with an imperfective infinitive, cf. *tebe*_{DAT.2SG} *vyxodit*_{INF-IMPERF} ‘You are coming out’, ‘You have to come out’, **tebe ne vyxodit*. Most Russian modal DPS predicatives can express both

¹⁶There are borderline cases like <Grustno>, *i nekomu*_{NEG-W1} *ne o čem*_{NEG-W1} *pisat* ‘<I am sad>, and I have no one and nothing to write about’, but generally Russian speakers avoid combining two *neg*-words without a conjunction. An anonymous reviewer points out that the assumption that all coordinate structures are bi-clausal is not shared by all linguists.

¹⁷Kondrashova & Šimík take for granted that the word *nem* is a contraction of **ne est*’.

alethic (external) or deontic/normative meanings (Zimmerling, 2020b). E.g., the sentence *mne*_{DAT.ISG} *nel'zja*_{PRED} *vyexat'*_{INF.PERF} has a natural alethic interpretation 'External Force prevents the speaker from leaving', but *mne*_{DAT.ISG} *nel'zja*_{PRED} *vyezžat'*_{INF.IMPERF} excludes such a reading and can mean either 'X is not allowed to leave' or 'X must stay here due to moral obligations.'¹⁸ Kondrashova (2008), Šimik (2009) and Kondrashova and Šimik (2013) hypothesize that the inbuilt modal component of MEC gets lost under the derivation of *neg*-words and cannot be spelled out.

2.3 Dative-infinitive structures and dative-predicative structures in Russian

DIS and DPS constructions have a prominent place in Russian grammar. The DIS construction, cf. (16b) above has the scheme $N_{\text{DAT}}-(\text{AUX})-\text{INF}$ and shares several features including the mechanism of dative case-marking with other Russian non-finite constructions with gerunds and participles as lexical heads. The earlier accounts, cf. Moore and Perlmutter (1999, 2000) and Zimmerling (2009) characterize this type of dative as *structural* since it is assigned uniformly by all infinitives and does not depend on the valency frame of the verb. However, in Marantz-style theories of case assignment, there are reasons to treat it as a *default case* of non-finite clauses in Russian (Zimmerling, 2020b): along the same lines, the nominative is chosen as the default case in finite clauses. Part of the discussion goes around the questions, of whether the silent subject of the non-finite predication (PRO) is case-marked in languages like Russian and Icelandic (Perlmutter & Moore, 2002) and whether overt dative subjects in DIS sentences like (16b) get their case from PRO or lend their case features to the embedded infinitival complement (Fleisher, 2006). The answer depends on the chosen theory: root DIS sentences can be described both as monoclausal (Švedova, 1982) and as bi-clausal, if one postulates a silent verbal head in the present indicative, cf. Fleisher's account (2006) accepted by Mitrenina (2017). Moore and Perlmutter (1999) treat the overt past and future tense markers *bylo*_{PST.3SG.N} and *budet*_{FUT.3SG} in DIS as temporal particles with a modal flavor: they cannot be replaced by auxiliaries *stat'* or *delat'sja*, which occur in standard Russian copular structures. The interaction of modal force and verbal aspect is discussed by Fortuin (2005, 2007). Israeli in a series of her publications (2013, 2014, 2016) focuses on non-indicative uses of DIS, notably in the contexts of the optative particle *by* and emphatic particle *že*.

The DPS construction is a finite sentence pattern with the scheme $N_{\text{DAT}}-\text{AUX}-\text{PRED}$. The dative case is assigned lexically by non-agreeing elements belonging to the class of dative predicatives (Zimmerling, 2009). Experimental studies confirm that many native speakers of Russian have 200–250 DPS items in their active vocabulary, while there is a considerable variation in the inventories of DPS items across the idiolects of different speakers (Zimmerling, 2021a, p. 251). All Russian DPS predicatives are stage-level predicates (SLP) that denote spatiotemporal situations and cannot be used as individual-level predicates (ILP): this constraint is captured already by Ščerba's pioneer work (1928). Overt realization of the dative subject is optional, but this also holds for dative subjects in DIS and

¹⁸In this pair of examples, the form of the perfective infinitive (*vyexat'*) contributes to selecting an alethic reading of *nel'zja* 'cannot' or 'must not' or 'it is forbidden'. However, the perfective form does not force an alethic reading and the imperfective form does not force a deontic/normative reading, since many contexts are ambiguous. E.g., the sentences *Emu*_{DAT.3SG.M} *nadopred* *bylo*_{AUX.PST.3SG.N} *uexat'*_{INF.PERF} / *uežžat'*_{INF.IMPERF} are compatible both with the reading 'External Force caused him to leave, and he left', cf. *Emu nado bylo uexat' / uežžat', i on uexal'*, and with the reading 'He had to/was obliged to leave, but he stayed', cf. *Emu nado bylo uexat' / uežžat', no on ostalsja*. A preliminary generalization over Russian DPS sentences is that the type of modality is not established in the embedded infinitival clause.

MEC sentences. A feature that opposes DPS to DIS is that DPS subjects are obligatory animate (+ ANIM), while DIS subjects are indiscriminate $\pm$ ANIM: a DIS sentence {*Gruzovikam*_{DAT.PL}}^{NON-ANIM} *ne*_{NEG} *proexat*'_{INF} 'The trucks cannot go through' is well-formed, but DPS sentences like {*Gruzovikam*_{DAT.PL}}^{NON-ANIM} *bylo*_{AUX.PST.SG.N} *grustno*_{PRED}, int. *'the trucks were sad', *{*Gruzovikam*_{DAT.PL}}^{NON-ANIM} *neprilično ne*_{NEG} *proexat*'_{INF} *zdes*', int. *'It is indecent for the trucks not to drive here' are ungrammatical (Zimmerling, 2020b). The ontology of the Russian DPS construction is broad and covers different kinds of spatiotemporal situations with an animate experiential subject, except for situations involving a symptomatic activity on the part of the subject: the last constraint blocks sentences like **mne zlobno*, int. 'I am furious' (Zimmerling, 2018a). The subcategorization of the DPS lexicon into thematic classes is a matter of debate, cf. Kustova (2021), Letuchiy (2022). Zimmerling (2021a, pp. 248–259) lists 15 classes, including modal predicatives *nel'zja*, *možno*, *nado*, *nužno*, *neobxodimo*, *nevozmožno*, etc., which express the basic deontic or alethic concepts.

The accounts, where all patterns with dative-case marking on the priority argument are considered variants of the same construction or explained by the same conditions on the derivation of impersonals (Moore and Perlmutter 1999, 2000; Perlmutter & Moore, 2002; Bonč-Osmolovskaja, 2003) explore the idea that language-specific constructions sharing pieces of structure (the DAT position + INF) share some general semantics. However, there are five clear-cut differences between DPS and DIS: 1) the dative case is assigned lexically in DPS and non-lexically in DIS; 2) the subject of DPS is obligatory + ANIM, while in DIS animacy is not a constraining factor; 3) DPS sentences are TPs with standard copulas and are adapted to all tenses, while DIS sentences have residual particle-like tense markers and are oriented towards present indicative, where the variety of modal meanings is maximal; 4) the modal meaning in DPS depends on the lexical head, while in DIS it is an inherent feature of the non-finite construction; 5) In DIS, the infinitive is obligatory, in DPS it is optional. A recent attempt to unify the syntax of DIS and DPS comes from Mitrenina (2017, pp. 66–68), who claims that all dative impersonals in Russian are headed by a silent operator with a combination of features EXIST + HAVE + [MODAL]: the meaning of DPS items like (*X-u*) *strašno* 'X is afraid', (*X-u*) *izvestno* 'X knows', 'It is known to X that' is decomposed 'X HAS fear/knowledge that P'. This does not explain, why some mental abilities and affections are linked with DPS predicatives, while others are not, cf. *nedoumenie* 'perplexity', **mne nedoumenno*, int. 'I am perplexed'.

The status of Russian *neg*-words regarding DIS and DPS sentences is even less certain. Feature 3) puts them together with DPS, while features 1) and 5) classify them with DIS. Feature 4) gives mixed results: *neg*-words project an idiomatic sentence pattern, while the meaning of external possibility can be expressed both by DIS and DPS sentences. Finally, feature 2) deserves further investigation. Most authors describe *neg*-words as + ANIM and rule out exceptions by special explanations. Apresjan and Iomdin (1989, p. 60) argue that non-animate subjects are exceptionally possible in set phrases like *jabloku*_{DAT.SG}^{NON-ANIM} *negde*_{NEG-W} *upast*'_{INF} 'This place is packed', lit. 'no place **for an apple** to fall', while Fortuin (2014, p. 36) claims that non-animate participants are only permitted if 'a tendency can be ascribed to the dative participant to realize the infinitive situation', i.e., if they are treated as active entities resembling animate subjects because of personification, metaphor or functional similarity. However, there are examples with underlying inactive non-animate subjects that escape either explanation, cf. (17) and (18).

(17) Russian [RNC]

Est' bog... bezborodyj, potomu **borode**_{DAT.SG.}^{NON-ANIM} *negde*_{NEG-W} na nëm pomestit'sja_{INF} [Sergei Klyčkov, 1926].

'There is God. He is beardless since the beard has nowhere to fit on him.'

- (18) Russian [RNC]
 V rešajuščie momenty **ětomu**_{DAT.SG} **straxu**_{DAT.SG}^{NON-ANIM} negde_{NEG-W}
 prosočit'sja_{INF}, v ètot moment čelovek dejstvuët. [Alexander Kron, 1983].
 'At crucial moments, this fear has nowhere to seep in, and at this moment the person acts.'

The Russian DPS construction has external parallels in other languages including Ukrainian, Bulgarian, Macedonian, Slovenian, Bosnian-Croatian-Serbian and Old Icelandic (Zimmerling, 2018a), while DIS construction is characteristic of Old Slavic languages including Old Church Slavonic (Khodova, 1980) and Old Russian but gradually eliminated except for the East Slavic group. That makes the preservation of two productive dative patterns an unusual feature of Modern Russian.

2.4 Diachronic studies

There are few studies devoted to the history of *neg*-words. Khodova (1980, p. 239) states that while Old Church Slavonic had a class of predicatives, the translators did not use contact sequences *ně* + *k*-word in negative MECs. Old Church Slavonic words consisting of *ně*- and *k*-words are used as indefinites, not as existential predicates so that OCS *něk*"*dě* conveys the meaning 'somewhere', but not 'there is no such place that' (Tseitlina et al., 1994, p. 386). Penkova and Rabus (2021, p. 228) point out that *ně*- is only one indefinite marker that survived in Modern Russian words with stressed initial *ne*-, cf. *nekto* 'someone', *nečto* 'something', *nekij* 'certain', also *neskol*'ko 'several' and *nekotoryj* 'some'. They outline a semantic map for Old Russian indefinites, classify the data according to the referential status expressed by four series of indefinites—affix-type *-ljubo*, *eter*- and *ně* items and zero-marked forms (Penkova and Rabus, 2021, pp. 236–242), and consider both contact and distant *ně* + *k*-word combinations, cf. (19):

- (19) Old Russian (Penkova and Rabus, 2021, p. 236)
 I **ně**_{NEG} bě_{IMPF.3SG} emou_{DAT.3SG.M} **kogowh**-ACC poslati_{INF}. (Galician chronicle [Hypat], l. 275 v. [1254]).
 'And he had no one to send at his disposal.'

Indefinite pronouns are considered underspecified items denoting sets of individuals, their categorial meaning being neither explicated by roots nor affixes (Kratzer & Shimoyama, 2002). This hypothesis predicts that the type of existential quantification can shift (Haspelmath, 1997), which holds for Old Russian *ně*- items. Synchronic accounts of Russian *neg*-words contain a conjecture that stressed *ne*- (** < ně*) in predicative *neg*-words consists of unstressed bare negation *ne* + *est'*_{BE.3SG}. This claim is absent from historical grammars: if *ně*- really contains a 3rd person singular BE-auxiliary, it can only be Proto-Slavic **-je*, not **(j)estī*. Likewise, the Modern Russian predicative negation *hem* is a contraction of **ne je tou* 'is not here', not **ne estī tou*. The form **-je* is not a late contraction of **jesti*, but a remnant of the Indo-European series of 3rd person forms without the ending **-ti*, so-called injunctive (Žolobov, 2016). The IE injunctive is presumably associated with non-indicative contexts. The distribution of Old Russian verb forms with and without **-ti* is a matter of debate, but if the injunctive semantics is preserved in Old Russian, the affirmative marker *est*' cannot be a component of the *neg*-word.

Several works discuss the evolution of DIS sentences. Madariaga (2011, p. 304) argues that infinitival complements were non-control structures in Old Russian, which made embedded DIS sentences with overt subjects possible, while Modern Russian generalized obligatory control in infinitival complements, therefore overt subjects are banned in this type of

clauses. This change is triggered by a shift in the settings of the *pro*-drop parameter and catalyzed by the loss of the overt tense marker in the present indicative. She implements a movement theory of control and assumes that with obligatory control non-finite subjects raise to the matrix clause leaving an NP-trace in the complement clause. Gerasimova (2015, p. 57) outlines a different approach to infinitival clauses in Modern Russian and argues that experiential predicates display arbitrary control in the absence of overt oblique subjects: according to her, infinitival complements have different status with subject (VP), object (TP), and arbitrary control (CP). If this is correct, the elimination of embedded DIS sentences in Modern Russian can be alternatively explained by the restructuring of CP or TP-infinitives. Zimmerling (2020b, 2021a, pp. 304–306) provides several examples of embedded DIS clauses in Russian and treats them as relics of Old Russian syntax.

2.5 State of the art: interim summary

Russian *neg*-words have been extensively analyzed as modal existential operators: their features are generally valued on a class of constructions with embedded *wh*-complements. For most authors, the analyses of *neg*-words as syntactic structures and as predicatives are mutually exclusive. The *neg*-words project an idiomatic sentence pattern and its comparison with two major sentence patterns characteristic of Russian grammar—Dative-infinitive structures (DIS) and Dative-predicative structures (DPS) is essential for checking such features as animacy, embedding and raising properties.

3 Internal structure and derivation of Russian *neg*-words

In this section, I outline a new account of the internal syntax of Russian *neg*-words. I first show that *neg*-words lack the animacy condition and argue that inanimate subjects are raised. Second, I argue that *k*-words lose morphological case if they raise to non-argument positions but retain case if they take the dative subject position, which is only possible for *nekomu* and *nečemu*. These facts support the view that the final interpretation of *neg*-words is established in the process of derivation, not in the lexicon.

3.1 Hybrid and degraded *neg*-words

I introduce the distinction of hybrid and degraded *neg*-words. A *neg*-word and the clause where it is realized is called *hybrid* [MEC predicate] if it combines the modal and the existential components and has a natural language negation. If the existential component is absent and the sentence cannot be negated, the *neg*-word is called *degraded*. The degraded uses of *neg*-words are subcategorized into *modal* and *derived*. The former express some sort of non-existential modality, cf. *nečego*² ‘one should not’, the latter express non-modal non-existential meanings, typically—the meaning of internal state, cf. *mne seččas nekogda* ‘I am busy right now’, **mne seččas est’ kogda*. With *nečego*, there is a pair of lexical homonyms: hybrid *nečego*¹ + Inf, where the *k*-word gets the morphological case from the verb, cf. *nečego*¹ *pit’*_{INF} ‘nothing to drink’¹⁹ and the modal predicative *nečego*², cf. *nečego*² + *xvalit’*_{INF} ‘should not praise someone’.

¹⁹The genitive case form in *nečego*¹ *pit’* is a standard modification of the accusative object *pit’ čto-to*_{ACC} in the context of negation.

(20) Russian

- a. **Nečego²**_{PRED} (tebe_{DAT.2SG}) xvalit'_{INF} studentov_{GEN-ACC.PL}.
'You should not praise the students.'
- b. ***Est'** (tebe) čego xvalit' studentov.
Int. 'You have a reason to praise the students.'

(20a) is not a MEC sentence. The ill-formedness of (20b) is due to the absence of the existential operator in (20a).²⁰ The degraded *neg*-word *nečego²* lacks case: the object slot of the verb *xvalit'* is taken by the NP *studentov*, not by the *k*-word *čego*.²¹ That means that *nečego²* is reanalyzed as a modal predicative.²² Apresjan and Iomdin (2010, pp. 68, 104) mention *nekogda* 'X is short of time' and argue that it is reanalyzed as a predicative denoting internal state. In many contexts, *nekogda* patterns with hybrid MEC predicates, but some of its uses are degraded, cf. (21).

(21) Russian

- a. (Mne_{DAT.1SG}) sejčas nekogda_{PRED} polot'_{INF} grjadku_{ACC.SG}.
'I have no time just now to weed the garden bed.'
- b. *(Mne_{DAT.1SG}) sejčas **est'** kogda polot' grjadku.
Int. 'I have time to weed the garden bed just now.'

The *neg*-words *ne k čemu* 'no need', 'there is no <internal> motivation for *p*' and *nezačem* 'no need', 'not for any purpose' mostly pattern with hybrids, cf. (22a–b) but express internal modality and not the meaning of external impossibility predicted by the MEC pattern.

(22) Russian

- a. I emu_{DAT.3.SG.M} bylo_{AUX.PST.3SG.N} **nekogda**_{NEG-W1} i **ne k čemu**_{NEG-W2} vmešivat'sja_{INF} v ljudskuju suetu vokrug. [RNC, Evgeni Schwarz, 1956].
'And he had **no time and no need** to interfere in the hustle and bustle around him.'
- b. Emu **bylo** kogda i k čemu vmešivat'sja v ètu suetu.
'He had both time and need to interfere in the hustle and bustle.'

In present-day Russian, the modal *neg*-word *nezačem* 'no need' is a homonym of the prepositional instrumental form *ne za čemu*_{INSTR.PREP}, lit. 'not behind what' with the *k*-word marked by INSTR. In the pair modal *ne k čemu²* 'no need', e.g., *ne k čemu² vozvraščat'sja* 'no need to return' vs hybrid MEC *ne k čemu_{DAT}¹*, e.g., *ne k čemu¹ pridrat'sja* 'nothing to find fault with' the spelling does not prompt that these are homonyms, but such a description is possible: from 150 RNC examples with an overt dative subject, the ratio of modal *ne k čemu²* is 68,7% (103/150). The drift from external to internal modality is probably due to the influence of the DPS construction since DIS sentences do not express this sort of modals. With

²⁰ An anonymous reviewer wonders whether the ungrammaticality of (21b) could be alternatively explained by the absence of the head assigning GEN to *čego*. This explanation does not work, since the *k*-word *čego* is licensed in the affirmative MEC sentences, cf. *est' čego prokrutit'* 'there is something to twist' (RNC, 2007), *est' čego axat'*, lit. 'there is something to groan / gasp at' (RNC, 1880). The same holds for *wh*-questions, cf. *Čego tut axat'* 'Why groan here?'

²¹ For the discussion of case loss see Sect. 3.4. below. Note that the frozen genitive ending inherited from the *k*-word *čego* is present both in *nečego¹* and in *nečego²*.

²² The lexical homonymy of the hybrid *nečego¹* vs. modal *nečego²* is easier to explain in the accounts, where *neg*-words are formed before syntactic derivation, cf. Rappaport (1986) or the Meaning-Text-Theory. With the derivational analysis adopted in this paper one can assume that the modal *nečego²* is merged higher than the hybrid *nečego¹*. I am grateful to the anonymous reviewer for the suggestion.

other *neg*-words, modal uses are more characteristic for the language of the XIX—first half of the XX century than for present-day Russian. The example (23), where the caseless form *ne iz čego*², lit. ‘not from what’ conveys the meaning ‘not for any reason’, sounds archaic.

- (23) Russian [RNC]
 Voevat'_{INF} nam_{DAT.IPL} s vami_{INSTR.PREP} **ne iz čego**²_{PRED}. (Mikhail Šolokhov, 1928–1940).
 ‘There is no reason that we and you are at war.’, lit. ‘there is not for anything for us to be at war with you.’

The uses of *neg*-words in contexts where no infinitival complement is possible lack natural language negation and are degraded. This holds for all contexts, where *neg*-words select nominative subjects.

- (24) Russian [RNC]
 Dlja dočki_{GEN.PREP} **tože knigi**_{NOM.PL} **nezačem**_{PRED}, potomu èto ne ženskoe delo. (Nicolai Leikin, 1879–1898).
 ‘<My> daughter **does not need books either**, because it is not a woman’s business.’

Similar side-uses with nominative subjects are attested by DPS predicatives *ne s ruki* ‘inconvenient for someone’, *ne pod silu* ‘beyond one’s reach’, in earlier Russian also *nevdoměk* ‘*X* cannot make it out’.²³ The data from RNC shows that while a majority of *neg*-words patterns with negative MECs, the deviations from the expected meaning of external impossibility are regular: all degraded *neg*-words and some hybrid *neg*-words, cf. (22) and (23) lack it. Derived degraded and modal meanings crucially rely on the presence of an overt or covert + ANIM subject. Indeed, the internalization of the MEC meaning ‘there is not a single reason to laugh at *p*’ ⇒ ‘you should not laugh at *p*’ expressed by *nečemu* requires a context, where some person can be reproached for laughing, cf. (25).

- (25) Russian [RNC]
Xoxotat'_{INF} vovse **nečemu**_{NEG-W}; gorazdo lučše zabavljat'sja s det'mi, neželi delat' to, čto vy vse, gospoda voennye (Alexander Griboedov, 1817).
 ‘**There is absolutely nothing to laugh at**. It is much better to have fun with children than to do what all of you do, gentlemen of the military.’²⁴

The correlation between animacy and derived meanings displayed by Russian *neg*-words has not been mentioned before, because most linguists take for granted that Russian *neg*-words invariably select + ANIM subjects. This issue is checked below in Sect. 3.3.

3.2 Inventory of *neg*-words

The class of *neg*-words is strictly limited, even if one counts all prepositional forms on a separate basis. Unlike Russian *koe*-indefinites, *ni*- negatives and reciprocals, which occasionally license splitting by secondary prepositions like *blagodarja* ‘thanks to’ (Arkadiev,

²³Cf. a late example: *Raz ètomu oluxu*_{DAT.SG.M} **nevdomek**_{PRED} *raznica*_{NOM.SG.F} *meždu Zemlej i asteroidom...* ‘Since this idiot cannot make out the difference between Earth and an asteroid...’ (RNC, 1967).

²⁴For those speakers of Modern Russian, who lack the government pattern *smejat'sja*, *xoxotat' čemu-l.*_{DAT} ‘to smile, laugh at something’ with a bare dative NP instead of the prepositional instrumental, *xoxotat' nad čem-l.*, sentences like (26) are both degraded and modal, since these speakers are incapable of retrieving the affirmative counterpart ²⁵*Est' čemu smejať'sja*, while in the XIX century Russian this pattern was still in use so that *nečemu* in (26) in this period was modal but not degraded.

Table 1 *Neg*-words in the Russian National Corpus (date of access: 24.08.2022)

Bare forms		NE + ČTO	NE + KTO
GEN-ACC		<i>nečego</i> ¹⁻² 38875*	<i>nekogo</i> 1679
DAT		<i>nečemu</i> 761	<i>nekomu</i> 6196*
INSTR		<i>nečem</i> 227	<i>nekem</i> 227
Case	Preposition	NE + ČTO	NE + KTO
GEN	<i>dl'ja</i>	<i>ne dl'ja čego</i> 565	<i>ne d'ja kogo</i> 139
	<i>di</i>	<i>ne do čego</i> 3	<i>ne do kogo</i> 13
	<i>iz</i>	<i>ne iz čego</i> 464	<i>ne iz kogo</i> 39
	<i>iz-za</i>	<i>ne iz-za čego</i> 110	<i>ne iz-za kogo</i> 3
	<i>ot</i>	<i>ne ot čego</i> 122	<i>ne ot kogo</i> 155
	<i>u</i>	<i>ne u čego</i> 24	<i>ne u kogo</i> 551
DAT	<i>k</i>	<i>ne k čemu</i> 737	<i>ne k komu</i> 370
	<i>po</i>	<i>ne po čemu</i> 23	<i>ne po komu</i> 1
ACC	<i>v</i>	<i>ne v čto</i> 2	<i>ne v kogo</i> 44
	<i>vo</i>	<i>ne vo čto</i> 202	
	<i>za</i>	<i>ne za čto</i> 1640	<i>ne za kogo</i> 63
	<i>na + čego</i>	<i>ne na čego</i> 2	<i>ne na kogo</i> 504
	<i>na + čto</i>	<i>ne na čto</i> 1913	
	<i>o(ob,obo)</i>	<i>ne o čto</i> 1	#
		<i>ne obo čto</i> 9	
	<i>pod</i>	<i>ne pod čego</i> 0	<i>ne pod kogo</i> 1
	<i>pro</i>	<i>ne pro čto</i> 3	<i>ne pro kogo</i> 1
	<i>s</i>	<i>ne s čego</i> 151	<i>ne s kogo</i> 113
	<i>čerez</i>	<i>ne čerez čto</i> 1	<i>ne čerez kogo</i> 5
INSTR	<i>za</i>	<i>ne za čem</i> 2	<i>ne za kem</i> 25
	<i>nad</i>	<i>ne nad čem</i> 84	<i>ne nad kem</i> 1
	<i>pered</i>	<i>ne pered čem</i> 2	<i>ne pered kem</i> 106
	<i>pod</i>	<i>ne pod čem</i> 2	<i>ne pod kem</i> 0
	<i>s</i>	<i>ne s čem</i> 386	<i>ne s kem</i> 1769
LOC	<i>v</i>	<i>ne v čem</i> 644	<i>ne v kom</i> 3
	<i>na</i>	<i>ne na čem</i> 651	<i>ne na kom</i> 50
	<i>o (ob, obo)</i>	<i>ne o čem</i> 2201	<i>ne o kom</i> 41
		<i>ne ob čem</i> 84	<i>ne ob kom</i> 1
	<i>po</i>	<i>ne po čem</i> 12	<i>ne po kom</i> 3
	<i>pri</i>	<i>ne pri čem</i> 2	<i>ne pri kom</i> 1
Adverbials	<i>nekogda</i> 20256*, <i>nekuda</i> 8967*, <i>negde</i> 3999, <i>nezačem</i> 3447, <i>neotkuda</i> 781		

2016; Zimmerling, 2018b), the *neg*-words can be split by primary prepositions only. RNC shows 62 *neg*-words, which is slightly below their full combinatorics: the forms *ne iz-pod čego* and *ne iz pod kogo* are not attested, since they call for special contexts. Table 1 provides the number of hits for each *neg*-word in RNC;²⁵ in all cases except for *nečego*¹⁻², *nekomu*, *nekuda*, *negde*, where the sample was overlarge, the data was filtered out by an annotator. The asterisk after the figure, cf. '*nečego*¹⁻² 38875*' indicates that *nečego* occurred 38875 times, and the overlarge sample was not annotated syntactically and semantically. The tag '#' stands for a morphological gap: the combination **ne o kto* does not exist. The absence of hits for the combinations *ne pod čego* (0) and *ne pod kem* (0) reflects the shape of the corpus by the date of access and captures the fact that the corresponding *neg*-words require special contexts which are not expected to occur often.²⁶

Table 1 shows that the forms of the inanimate series *NE + ČTO* are generally more frequent compared to forms from the animate series *NE + KTO*, except for the combinations

²⁵Russian National Corpus: <https://ruscorpora.ru/> [Date of access: 24.08.2022].

²⁶The processing of ipm values for all Russian *neg*-words including the low frequent ones is a feasible but difficult task that requires a larger corpus than RNC.

ne + *u* + *k-word* and *ne* + *s* + *k-word*, where animate *neg*-words are more prominent. The forms of the inanimate series can have modal or degraded uses but to retrieve it from RNC, semantic annotation is needed.

3.3 Animacy, control and raising

The animacy restriction does not hold with *neg*-words. The proof comes from sentences where the subject N_{DAT} slot is filled by an NP with the features ‘lexical inanimacy + inactivity’ and no set phrase is formed. Such sentences are attested with most *neg*-words. The statistics in Table 2 show the total number of RNC hits for four *neg*-words—*ne na čem*, *neotkuda*, *nekuda* and *negde*—in clauses with overt dative subjects. The samples have been annotated syntactically, to tell the relevant examples, e.g., *n’e na čem* from homonymic uses, e.g., *ne na čem ino ‘m* and semantically, to classify their subjects as + ANIM or NON-ANIM. Ambivalent contexts, where the verb does not discriminate animacy, cf. *neotkuda vžjat’ šja kazakam^{ANIM}/kustam^{NON-ANIM}* ‘nowhere to take the Cossacks/ the bushes from’ or part of the body can be analyzed as pars pro toto references to a human being, cf. *ne na čem glazu^{NON-ANIM} ostanovit’ šja* ‘no place for the eye to stop at’ are classified in Table 2 with the group ‘non-experiential inactive subjects’. The contexts, where NPs like *pravitel’s tvo* ‘government’ denoting groups of people combine with active predicates specified as + ANIM, cf. *inače pravitel’s tvo ne na čem budet stroit’ bjudžet* ‘otherwise the government will have nothing to build a budget on’ have been classified as having underlying animate subjects.

I claim that inanimate and other non-experiential inactive subjects of *neg*-words are raised internal arguments of *k*-words. The standard diagnostics for raising structures relies on the placement of adjuncts, scope, preservation of idiom chunks and passivization (Burukina, 2019), but the validity of these tests in a language with non-restrictive word order needs to be checked in the experiment. Lyutikova (2022a, p. 40) claims that Russian phasal and modal verbs have raising syntax but undergo restructuring, while Letuchiy and Viklova (2020, p. 57) classify Russian constructions into four groups, with two intermediate types—sentences with a hidden restrictor and ECM-type sentences—placed between proper control structures with volitive verbs and raising structures with *okazat’ šja* ‘turn out’, *stat’* ‘become’ and *sčitat’*

Table 2 Inanimate dative subjects with *neg*-words in RNC (date of access: 24.08.2022)

Neg-word	All sentences with overt dative subjects	All non-experiential inactive subjects	Inanimate inactive subjects
<i>ne na čem</i>	178 (100%)	77 (43,2%)	18 (10,1%)
<i>neotkuda</i>	346 (100%)	140 (40, 4%)	125 (36,1%)
<i>nekuda</i>	253 (100%) ²⁷	49 (19,4%) ²⁸	36 (14, 2%)
<i>negde</i>	139 (100%) ²⁹	122 (87,8%) ³⁰	16 (11,5%)

²⁷For a sample of 253 RNC sentences retrieved by the search query ‘*nekuda* + noun in the dative case in the contact position <−1; 1>’. The query returned 400 sentences, 147 of which were filtered out manually. The hypothesis that the search items are the same clause elements assuming a subject-predicate relation returned true in 58%.

²⁸With the collocation *jabloku nekuda upast’* ‘there is no place for an apple to fall’.

²⁹For a sample of 139 RNC sentences retrieved by the search query for *negde* and a noun in the dative case in the contact position <−1; 1>. The query returned 218 sentences, 79 of which were filtered out manually. The hypothesis that the search items are the same clause elements assuming a subject-predicate relation returned true in 63,7%.

³⁰The abnormally high ratio of inanimate subjects for *negde* is explained by two factors: 1) high frequent collocation *jabloku negde upast’* ‘the apple has nowhere to fall’, where *jabloko* normally assumes a contact position with *negde*; 2) animate subjects mostly take a distant position with *negde*.

‘consider’. In this paper, I tentatively assume that the control vs raising contrast is sufficiently captured in Russian by two criteria: 1) control structures impose selectional restrictions on the feature specification of the matrix predicate and/or its arguments, e.g., the animacy conditions, while raising structures do not; 2) raising structures license narrow scope interpretation, while control structures exclude it. I also assume that narrow scope reading forces an alethic reading of the modal predicate, cf. Zimmerling (2020a). If one and the same Russian *neg*-word like Russian DPS predicatives of the type *nado* ‘must’ or ‘need’ combines with two kinds of dative subjects,—animate experiential arguments, cf. *mne*_{DAT.SG}^{ANIM} *nado*_{PRED} *uexat*_{INF} ‘I must leave’, *mne*_{DAT.SG}^{ANIM} *nado*_{PRED} *ostyt*_{INF} *posle trenirovki* ‘I need to cool off after training’ and non-animate arguments with the role of Theme or other inactive roles, cf. *pirogu*_{DAT.SG}^{NON-ANIM} *nado*_{PRED} *ostyt*_{INF} ‘the pie needs to cool down’, the latter can be analyzed as raised internal arguments. The support for this hypothesis comes from the fact that raised structures with the modal DPS predicatives have a fixed alethic reading: a pie that needs to cool down, unlike a human cannot be the subject of any deontic or normative attitude. With *neg*-words, the presumably raised internal argument belongs to the *k*-word, not the infinitive, as with DPS predicatives. Nevertheless, the raising analysis is possible. (26) contains the copula *okazat’sja* ‘turn out’, which is a raising verb (Lyutikova, 2022b, p. 61).³¹

(26) Russian [RNC]

I **mnogoobraziju**_{DAT.SG} **znanij o prošlom** *okazalos*_{AUX.PST.3SG.N} *ne na čem*_{NEG-W} *deržat*_{’sja}_{INF}. (Olga Balla, 2005)
 ‘And it turned out that **the diversity of knowledge about the past** has nowhere to hold on to.’

It is not straightforward, whether obligatory animate subjects must be explained the same way. The silent subject of a *neg*-word is always interpreted as + ANIM, cf. (27a–b). This can be due to the fact that it is a raised external argument of the *k*-word, but this explanation seems to hang on the assumption that clauses with *neg*-word have obligatory control, which lacks an independent verification. *Wh*-questions like (27c–d) are weird, which prompts that structures like (27a) have arbitrary control.³²

(27) Russian

- a. Ø_i^{ANIM} *Neotkuda*_{NEG-W} [*PRO*_i *posylat*_{INF} *den’gi*_{ACC} *žene*_{DAT}].
 ‘Nowhere to send money from to one’s wife.’
- b. *Ivanu*_{DAT}^{ANIM} *neotkuda*_{NEG-W} [*PRO*_i *posylat*_{INF} *den’gi*_{ACC} *žene*_{DAT}].
 ‘Ivan has nowhere to send money from to his wife.’
- c. ?*Komu*_{WH-DAT} *posylat*_{INF} *den’gi* *žene*?
- d. ??*Komu*_{WH-DAT} *otkuda*_{NEG-W} *posylat*_{INF} *den’gi* *žene*?

Unless the obligatory control hypothesis is taken, animate dative subjects in sentences like (27b) can be regarded external arguments of *neg*-words, not infinitives or *k*-words. This is possible, if *neg*-words are not formed in complement clauses.³³ I adopt the raising analysis of *k*-words and postulate two slots for overt dative subjects, one in the complement clause and one in the matrix clause.

³¹ A possible objection is that the raising verbs *naftis*’, *okazat’sja* or *stat*’ do not discriminate animacy of the embedded clause arguments, cf. *Emu*_{DAT.3SG}^{ANIM} *okazalos*_{AUX.PST.3SG.N} *nekuda*_{NEG-W} *klast*_{INF} *sumku* ‘It turned out that he had to place to put the bag’. It is therefore better to explain the raising of the inanimate dative arguments by the internal syntax of Russian *neg*-words, and not by the type of the copula they combine.

³² Cf. a similar proposal by Gerasimova (2015).

³³ An anonymous reviewer provides another argument against treating obligatory animate dative subjects as external arguments of *neg*-words: the negation takes the proposition as its argument, not an individual.

(vi) [DAT1 [+ANIM], NEG, *k*-word_i, [DAT2 [± ANIM], INF, ~~*k*-word_i~~]].

I assume that inanimate subjects originate in embedded clauses and are raised, while animate experiential subjects are matrix clause elements merged after a *neg*-word is formed. A combination of the higher + ANIM dative subject with the raised lower ± ANIM dative is ruled out, cf. (28a), while the variant with a non-subject + ANIM prepositional genitive phrase³⁴ is fine, cf. (28b).

(28) Russian

- a. ***Mne**_{DAT}^{ANIM} *zdes'* *neotkuda*_{NEG-W} *vzjat'*_{sja}_{INF} **derev'jam**_{DAT}^{NON-ANIM} / **elektrikam**_{DAT}^{ANIM}.
Int. 'Here I have nowhere to take trees / electricians from at my location.'
- b. **U menja**_{GEN.PREP}^{ANIM} *zdes'* *neotkuda*_{NEG-W} *vzjat'*_{sja}_{INF} **derevjam/ elektrikam**.

3.4 Raising of *k*-words and case

Most authors assume that DPS predicatives lack agreement and case: this hypothesis was first raised by Pospelov (1955). Letuchiy (2018b) however argues that adjectival forms of the predicative instrumental in contexts like *ja*_{1SG} *sčitaju*_{PRES.1SG} *nužnym*_{PRED.INSTR} 'I find something necessary' can be linked with predicatives like (*X*-*u*) *nužno*_{PRED} '*X* needs' but other authors reject the idea that DPS items are ever used as secondary predicates (Zimmerling, 2021b). The analysis of BE-auxiliaries in DPS sentences as tense particles does not make sense since the AUX slot in DPS can be filled by other copulas, incl. *stat'* and *delat' sja*.

Degraded *neg*-words like *nečego*² 'should not' in (21) and all adverbial *neg*-words lack case. The status of argument hybrid *neg*-words is less certain since the case of the *k*-word is assigned by the verb. I nevertheless argue that all Russian *neg*-words lack morphological case and oblique case forms of *k*-words *nekto* and *nečto* lose it in the process of derivation, if they merge with the general negation in the predicate position. There is twofold motivation for this analysis. First, *neg*-words are similar to DPS predicatives all of which lack case even if they originate from frozen case forms like the old nominatives *byla*_{AUX.PST.3SG.F} *pora*_{NOM.SG.F}, *neoxota*_{NOM.SG.F} ⇒ *byla*_{AUX.PST.3SG.N} *pora*_{PRED}, *neoxota*_{PRED}. An even more instructive parallel is provided by the predicatives consisting of a frozen oblique case form attached to a prepositional/prefixal head, cf. the DPS predicative *emy*_{DAT.3SG.M} *было*_{AUX.PST.3SG.N} *bez raznicy*_{PRED} '*X* did not care' and the nominative predicative *On*_{NOM.3SG.M} *был*_{AUX.PST.3SG.N} *navesele*_{PRED} '*X* was tipsy', lit. '*X* was on the merry <side>'. The historical oblique case ending, cf. the well-formed PP *bez bol'soj*_{GEN.SG.F} *raznicy*_{GEN.SG.F} and the extinct PP **na vesele*_{LOC.SG.N} is visible, but since the prepositions/prefixes *bez* and *na* belonging to the predicative idioms *mne bez raznicy* (**mne bez bol'soj raznicy*) and *ja navesele* are no longer

I assume that sentences with Russian *neg*-words undergo restructuring, with reanalysis of the complex *ne*² + *k*-word as a new syntactic head NEG-W⁰. In terms of dependency grammar, the higher dative (DAT1) is the dependency of the NEG-W⁰ head. In terms of generalized phrase structure grammar, it can be located at SpecNP, though this step needs justification.

³⁴Note that the restriction on the simultaneous filling of DAT1 and DAT2 shown in (28a) holds because both dative arguments are subject-like, while the prepositional genitive *u menja* in (28b) is not. There are numerous proposals for postulating genitive and accusative subjects, such as the *u* + *GEN* construction for Russian. In this paper, I treat 'subject' as a structural, not as a functional or semantic notion. In sentences like *U Sergeja*_{GEN.PREP} *byl*_{AUX.PST.3SG.M} *gripp*_{NOM.SG.M} 'S. had flu' the nominative NP controls the number and gender form of the predicate, and I do not see an obvious way to treat this NP as nominative object, not as subject, since nominative objects are banned in Standard Russian, cf. **mne*_{DAT.1SG} *nužno bylo*_{AUX.PST.3SG.N} [_{NP} *lekarstva ot gripa*]_{NOM.PL}.

recognized as heads in syntax, one can neither treat the predicative idiom nor its part as case expressions. Second, if *neg*-words are morphological units, their case must be valued in the matrix clause. I propose that *k*-words, e.g., *kogo* ‘whom’ occur in hybrid structures like (29) twice: in the lower clause they take an argument position and are case-marked but lose case after raising to a non-argument position and incorporating into the NEG head in the higher clause.

- (29) $\emptyset_I$ ne^0 **kogo**_{PRED.NEG-W} [PRO_i vinit’_{INF} v krizise **kogo**_{ACC}].
 ‘There is no one to blame for the crisis.’

I admit that the case ending of the *k*-word in (29) is visible but claim that in the higher copy of *nekogo* the case feature is deactivated, i.e., ‘lost’ due to the fact that the *k*-word and the negation make up a morphological unit. Case deactivation can be similar to or different from case assignment. Case loss is not triggered by the raising in itself but is an effect of merging *ne*² with the *k*-word in the predicate position. However, this merger is fed up by the raising of *k*-words out of embedded clauses.

To prove that no predicative *neg*-word has case, one must find sentences, where the object slot of the infinitive is filled by another expression in the same case form and the same type of semantic argument as the *k*-word. This is a difficult task, since Russian *neg*-words unlike *ni*-negatives do not take non-predicative complements and do not incorporate full *wh*-phrases (**ne kakaja devočka*). However, RNC provides several examples with the words *drugoj*, *drugoe* ‘other’, *inoj*, *inoe* ‘someone else, something else’ that match this test.

- (30) Russian [RNC]

- a. Imenno tol’ko po odnomu tomu, čto **drugogo**_{ACC} nekogo_{ACC} podobrat’_{INF}!
 (Fedor Dostoevskij, 1880)
 ‘It is only because there is no **one else** to pick up.’
- b. podumal Oblomov, s užasom povodja glazami po stenam i ostanavlivaja ix na nosu Anis’i, potomu čto **na drugom**_{LOC} ostanovit’ ix bylo ne na čem_{LOC} (Ivan Gončarov, 1859)
 ‘...Oblomov thought rolling his eyes around the walls in horror and stopping them with Anisya’s nose, because it was impossible to stop them **on anything else**.’

Examples like (30a–b) are rare but grammatical. Note that it is impossible to reconstruct *neg*-phrases **nekogo drugogo*, **ne na čem (na) drugom*, **ne za čto (za) drugoe* either in XIX century Russian or in the contemporary language.³⁵ Another piece of evidence comes from hybrid *neg*-words combined with nouns apparently filling the object slot initially occupied by the *k*-word, cf. (31).

- (31) Russian [RNC]

- tak kak iz ètogo vychodit, čto mne_{DAT.ISG} ~~ne na čem~~_{LOC} osnovyvat’_{INF} **na položenii**_{LOC}, pozvoljauščem avtoram vybirat’ ispolnitelej. (Vladimir Tel’akovskij, 1902)
 ‘as it follows from this, I have **nothing** to base on, since I lack **the regulation** which allows the authors to select the actors themselves.’, lit. ‘I have **nothing** to base on **on the regulation**...’

³⁵Russian *ni*-negatives are NPI items that must be licensed by an overt negative head like *nel’zja*: *nel’zja*⁰ *podobrat’*_{INF} [_{NIP} *nikogo drugogo*], *nel’zja*⁰ *ostanovit’* ix [_{NIP} *ni na čem drugom*].

Sentences like (31) are not welcome in the written form of the present-day standard Russian, but the fact that the speakers produce them hints that the case feature of *k*-words is weakened, if not lost. A modest generalization is that at least in some sentences the embedded infinitive takes a case-marked complement filling in the object slot originally occupied by the *k*-word at the start of derivation. This is attested both with all modal *neg*-words like *nečego*² in (21) lacking the existential force and occasionally with hybrid *neg*-words projecting the negative MEC construction, cf. (30a–b) and (31).

3.5 Subject *neg*-words and argument structure

The *neg*-words *nekomu* and *nečemu* incorporating bare dative forms of the *wh*-pronouns *kto* and *čto* have an extra position available: they can be realized both as predicate heads and as subjects of DIS sentences. To match this pattern, subject *neg*-words must be case-marked. This state-of-the-art observation can be reconciled with the raising analysis of predicative *neg*-words, if one assumes that *k*-words raise both to argument and non-argument positions.

- (vii) Russian *k*-words retain morphological case if they raise to argument positions but lose case if they raise to the non-argument position of the lexical predicate head.

The DIS construction does not discriminate animacy, but the inanimate *neg*-word *nečemu* in the subject position constrains the verb type. While *nekomu*^{ANIM} combines with all verbs, *nečemu*^{NON-ANIM} typically combines with inactive and active intransitive verbs.

(32) Russian

- a. **Nečemu**_{NEG-W.DAT}^{NON-ANIM} bylo_{AUX.PST.3SG.N} **goret'**_{INF} [INTR, – active].
'There was nothing to burn in the fire.'
- b. **Nekomu**_{NEG-W.DAT}^{ANIM} bylo_{AUX.PST.3SG.N} **tušit'**_{INF} [TR, + active] **koster**_{ACC}.
'There was no one to put out the fire.'

Semantically, DIS sentences like (32a–b) are standard negative MECs, which bring about existential quantification. They have affirmative counterparts *bylo čemu goret'*, *bylo komu tušit'*, which classifies them with hybrid structures. The puzzling thing is that they do not project the NEG-WH—INF pattern diagnostic for predicative *neg*-words, while the latter licenses degraded and modal forms deviating from the expected MEC meaning of external impossibility. It makes sense to range three types of dative subjects in sentences with *neg*-words according to the size of verb classes. Obligatory + ANIM subjects taking the higher DAT1 slot do not constrain the verb type: this holds both for *nekomu* and for referential dative NPs or pronouns. Raised internal arguments with the values – ANIM or ± ANIM take the lower DAT2 slot and select intransitive verbs. *Nečemu* mostly combines with intransitive verbs (92,5% from the RNC sample³⁶), but Russian authors sporadically use it with transitive and semi-transitive verbs not specified as + ANIM.

(33) Russian [RNC]

- a. **Nečemu**_{NEG-W.DAT}^{NON-ANIM} bylo_{AUX.PST.3SG.N} **oživit'**_{INF} **bor'bu**_{ACC} **čeloveka**_{ACC.SG.M} **v skorbi** [TR, + caus, ± Anim]; **nečemu**_{NEG-W.DAT}^{NON-ANIM} bylo_{AUX.PST.3SG.N} **utešit'**_{INF} **ego**_{ACC.SG.M} **v skorbi** [TR, + caus]. (Vladimir Odoevskij, 1839)
'There was nothing to revive the struggle of the man, nothing to comfort him in his grief.'

³⁶RNC has 158 DIS sentences with *nečemu* in the subject position, 146 of which have intransitive verbs. The sample with *nečemu* in the predicative position with DAT slot filled by another expression is smaller—100 sentences, all of which have + ANIM subjects and transitive verbs.

Table 3 Dative subjects and verb classes in Russian sentences with *neg*-words

	Type of subject	Syntax	Semantics	Verb classes
I	raised internal argument [± ANIM]	NEG ⁰ — DAT2 — INF ⁰ — <i>k</i> -word	MEC (external impossibility)	intransitives
II	<i>nečemu</i> [-ANIM]	DAT1 _{NEG-k-word} — INF ⁰		intransitives and transitives [-ANIM]
III	<i>nekomu</i> [+ANIM]	DAT1 _{NEG-k-word} — INF ⁰		all verbs
IV	referential animate subjects [+ANIM]	DAT1 — NEG- <i>k</i> -word ⁰ — INF	MEC or derived meanings	all verbs

- b. To est' ne suščestvoval sam predmet psixologičeskogo analiza—**na psixiku**_{ACC.PREP} **nečemu**_{NEG-W.DAT}^{NON-ANIM} bylo_{AUX.PST.3SG.N} **vlijat'**_{INF}.
[S-TR, ± Anim] (“Družba narodov”, 1999.09.15)
‘That is, the subject of psychoanalysis did not exist—there was nothing to influence the psyche.’

The correlations between subject and verb properties in different subtypes of Russian sentences with *neg*-words are summed up in Table 3.

It is unclear, which category the scale Type I > Type II > Type III > Type IV stands for—activity, animacy or semantic control. Anyway, DPS-like features increase towards the right edge of the scale. Type IV has the greatest similarity with DPS sentences: these are finite structures that require + ANIM subjects but do not constrain the type of the embedded verb (infinitive). Type I resembles embedded DIS structures, which were productive in Old Russian but are marginal in Modern Russian: the *neg*-word behaves as an external head into which the block [DAT2 INF] is embedded. The two intermediate types are root DIS sentences with *neg*-words in the subject positions. What differs them from other root DIS is non-reduced copular syntax in declaratives, cf. *nekomu budet žit' zdes'* ‘nobody will live here’, **Ivanu (ne) budet (ne) žit' zdes'*, int. ‘Ivan will not live here’.

If there are two dative *neg*-words by a single spelled-out predicate, their surface order is not fixed, but the inanimate *nečemu* cannot be selected as subject over the animate *nekomu*.³⁷ Given that the ‘one *neg*-word per clause’ constraint holds for Russian, cf. Sect. 2.2 above, sentences like (34a) are bi-clausal and have the structure (34b).

(34) Russian

- a. *Nekomu*_{NEG-W.DAT}^{+ANIM} i *nečemu*_{NEG-W.PRED}^{NON-ANIM} *učit'sja*_{INF}.
‘There is no one and nothing to learn.’
b. ([DIS NEG-W1^{+ANIM} *učit'sja*]) & ([Neg-WP Ø^{+ANIM} NEG-W2^{NON-ANIM} [PRO *učit'sja*]]).

It is tempting to treat gapping in (34b) as proof that both conjoined clauses have DIS syntax. Unfortunately, gapping is licensed in Russian even if the conjuncts sharing parts of VP have a different build-up, e.g., if one conjunct is a TP and another one is non-finite, cf. (35) and (36).

(35) Russian

- a. *Neumestno*_{PRED} *vešat'*_{INF} *prestupnikov*_{ACC} i *nekomu*_{NEG-W.DAT}^{+ANIM}.
‘It is **not appropriate** to hang criminals, and there is no one to do it.’
b. ([TP PRED [PRO_i *vešat'* *prestupnikov*]]) & ([DIS NEG-W^{+ANIM} *vešat'* *prestupnikov*]).

³⁷ The raising of *nekomu* to the subject position is blocked, if it is an internal argument of the (semi) transitive verb with a dative slot. Cf. *verit' komu-l. i čemu-l.* ‘to trust someone or something’: *Tut nekomu i nečemu verit'* ‘There is no one and nothing to trust here.’ (RNC, Alexander Zinoviev, 1995) = *Tut ((nekomu verit') & (nečemu verit'))*.

(36) Russian [RNC]

- a. Čto do Goracija, to posle Axmatovoj na èti lavry pretendovat'_{INF} u nas nekomu_{NEG-W.DAT} ^{+ANIM} i nel'zja_{PRED}. (Iosif Brodskiy, 1993)
'Concerning Horatius, after Akhmatova, we have no one to claim his laurels and this is **not allowed**.'
- b. ([_{DIS} NEG-W^{+ANIM} {T°} pretendovat' na èti lavry) & ([_{TP} PRED {T°} pretendovat' na èti lavry]).

One can add an overt tense head, e.g., past tense marker *bylo* to (35) and (36) without yielding ungrammaticality. This indicates that DIS sentences with the subject *neg*-word *nekomu* pattern with TPs in gapping configurations.

3.6 Neg-words: interim summary

The MEC semantics (existential quantification) depends on the presence of a *neg*-word in syntactic structure with an embedded infinitive, not on its end position. Both predicative *neg*-words selecting the idiosyncratic NEG-W—INF pattern and subject *neg*-words selecting the productive DIS pattern (DAT—INF) are derived by raising of *k*-words into the matrix clause. However, predicative *neg*-words incorporating case-marked *k*-words lose case in the process of derivation, while subject dative *neg*-words retain it. The final interpretation of a *neg*-word depends on its position in the matrix clause (argument vs non-argument position). The *ne* + *k*-word agglomerates have different features at the start and the end of derivation. Contrary to previous descriptions, *neg*-words regularly license raised non-animate dative subjects of intransitive verbs and sporadically do it with other verb types. This finding compromises the idea that obligatory + ANIM dative subjects originate in embedded clauses with *k*-words

Table 4 Types of Russian neg-words, their semantics and syntactic patterns

Type of the <i>neg</i> -word		Semantics	Affirmative MEC correlate	Reference construction
1. Hybrid predicates:		negative MEC	yes	DIS, DPS
1.1. <i>mne</i> [+ANIM] <i>nekogo vinit'</i> , <i>mne</i> [+ANIM] <i>nečego¹ delat'</i> .		negative MEC	yes	DIS, DPS
1.2. <i>stolbu</i> [NON-ANIM] <i>ne na čem stojat'</i> .		negative MEC	yes	DIS
1.3. (<i>mne</i> [+ANIM]) <i>na drugom ostanovit'</i> <i>vzgljad ne na čem</i> .		negative MEC	yes	DPS
2. Hybrid subjects:		negative MEC	yes	
2.1. <i>Nekomu</i> [+ANIM] <i>tušit' kostěr</i> .		negative MEC	yes	DPS
2.2. <i>Nečemu</i> [NON-ANIM] <i>goret' v kostre</i> .		negative MEC	yes	DIS
3. Degraded predicates:	3a. Modal:	deontic, normative or internal modality	no	DPS, Dative-nominative structure
	3.1. <i>Nečego² tebe</i> [+ANIM] <i>xvalit' studentov</i> .	normative modality	no	DPS
	3.2. Arch. <i>Dočke</i> [+ANIM] <i>knigi nezačem</i> .	internal modality	no	Dative-nominative structure
	3b. Derived	internal state	no	DPS
	3.3. <i>Mne</i> [+ANIM] <i>sejčas nekogda polot' grjadku</i> .			

and supports the alternative hypothesis that they are matrix clause elements merged in structures with arbitrary control. I introduced several tags for the uses of Russian *neg*-words. The relation of their subtypes to positive MEC sentences and diverse constructions is summed up in Table 4.

4 Diachrony and derivation: from embedded DIS sentences to modal DPS predicatives

In the previous section, I demonstrated that two subtypes of the NEG-W—INF construction have different syntax. The sentences with raised internal non-agentive arguments (*borode negde pomestit'sja, èpidemii neotkuda vzjat'sja, stolu negde vstat'*) are embedded DIS structures with an external head. The sentences with an overt or underlying + ANIM subject (*Mne^{ANIM} negde stat' ~ Ø^{ANIM} negde stat'*) involve a reanalysis of a *neg*-word as a DPS predicative selecting a matrix clause subject. What differentiates hybrid *neg*-words from true DPS predicatives is that the latter do not express existential quantification. However, the deviations from MEC semantics and the expected meaning of external impossibility are most regular in the context of + ANIM subjects, which indicates that this subtype of the NEG-W—INF construction is influenced by the DPS pattern with obligatory + ANIM experiential subjects. In the final section, I argue that this derivational perspective corresponds to the actual grammaticalization vector: NEG-W—INF structures originated as embedded DIS clauses but were reanalyzed as DPS clauses.

4.1 Embedded DIS clauses

Old Russian licensed embedded DIS clauses even with non-identical subjects, cf. (37) with a dative predicative *godě* in the matrix clause taking a DIS complement.

- (37) Old Russian (Madariaga, 2011, p. 315)
 Ašče godě_{PRED} ti_{DAT,2SG} est' [DIS žiti_{INF} nam"_{DATj} v" nem"] budi nam"_j pomoščnik (Theod., 81)
 'If it is convenient for you that we live in it, be our assistant.'

This type of DIS is ruled out in Modern Russian, which straightforwardly accounts for the ill-formedness of the combination of higher animate and lower inanimate dative subjects in MECs like **Mne zdes' neotkuda vzjat'sja derevjam*. However, if the dative slot in the higher clause is vacant, argument raising is possible not only by *neg*-words but by modal dative predicative *nado* 'is needed', cf. the raising structure (38a), which conveys the same meaning as its finite counterpart (38b).

- (38) Russian (Zimmerling, 2020b)
 a. Pirogu_{DAT} ^{NON-ANIM} nadop_{PRED} [DIS PRO ostyt'_{INF} pirogu_{DAT}].
 'We need the pie to cool down.'
 b. Ø Nadop_{PRED}, [CP čtoby pirog_{NOM} ostyl].
 'The same.'

Other modal DPS predicatives, e.g., *nužno*, *neobxodimo*, *nevozmožno*, *nel'zja* license embedded DIS clause like only marginally, cf. ?*pirogu nužno ostyt'*. Meanwhile, raising of dative arguments is acceptable in optative and comparative contexts with *xorošo* *by* 'it would be nice' and *lučše* '<it would be> better'. Cf. a fresh example: a noted football fan, artist N. (b. 1957) reacted to a disastrous play of the goalkeeper M. by a verse starting with (39a).

- (39) Russian (“Sport-express”, 25.10.2021³⁸)
- V vorota xorošo_{PRED} by_{OPT} stat’_{INF-PERF} golkiperu_{DAT}... <a ne mal’čiške vo vratarskom svitere s povadkami vnimatel’nogo zritel’a.>
‘It would be nice if there was a real goalkeeper in the goal, <and not that kid M.>.’
 - Golkiperu_{DAT} xorošo_{PRED} by_{OPT} [DIS stat’_{INF} v vorota golkiperu_{DAT}]

The context (39) excludes a matrix clause deontic reading ‘It is good for M. himself to be in the goal’: the speaker asserts a different thing—that it would be nice if the goalie were someone else, not M. Speaking of similar sentences in Old Russian, Borkovskij (1978, p. 256) claims that until the XVIII century, dative subjects were infinitival dependencies and claims that the modern DPS construction, where the dative subject depends on the predicative, results from a recent reanalysis of DIS clauses. This is an exaggeration, since DIS and DPS constructions develop in parallel, while for many contexts with Old Russian DPS predicatives it is impossible to offer any raising interpretation, cf. *žal’* ‘X is sorry’ taking a finite complement clause in (40). *Ně l’žě* ‘cannot’ taking an infinitival complement in (41) is ambiguous between DPS and DIS readings, but since the subject *mi* is in the first person, the DPS reading ‘I cannot do *p* right now’ is preferable to the DIS reading ‘It is impossible that *X* does *p*’.³⁹

- (40) Old Russian
Žal’_{PRED} bo ny_{DAT} est’_{AUX.PRES.3SG} bra^t_{ACC} našego_{ACC} deržiši_{PRES.2SG} Igorja_{ACC}.
‘Because we are very sorry that you hold our brother Igor.’ (Kyev chronicle [Hypat.], [1146])
- (41) Old Russian (Zimmerling, 2022)
I se emu javiši cr’ na mja Greckyi v”stavaet’ rat’ju. i seš mi_{DAT} zimy i vesny nē lzē_{PRED} na kon’ k tobě vsēsti_{INF}. no wbače wce tvoi ščit’ i moi ne rozno esta. ače mi_{DAT} samomu_{DAT} nē nlē_{PRED}. a pomoči koli xočeši. i”li tysjac’ bolša li a to ti poslju. (Kyev chronicle [Hypat.], [1150])
‘And <the king> declared him: “The Greek emperor attacks me now, and I cannot lead cavalry to you this winter and spring. But since, father, your shield and my shield stand together and you want to get help, even if I cannot <come> myself, I will send you 10000 soldiers or more”.’

The modest generalization over Modern and Old Russian data is that some sentences with modal predicatives are ambiguous between a DPS (=matrix clause) and DIS (=embedded clause) reading of dative subjects, but while this kind of ambiguity is frequent in Old Russian texts, it is only exceptionally preserved in Modern Russian with syntactic heads like *nado* and *xorošo by* in (39). (38a) is an unambiguous raising structure, since the DPS construction rules out inanimate subjects, but if one replaces *pirogu* with an animate dative NP, e.g., *vsadniku* ‘a rider’, the sentence would be two-way ambiguous. Predicative *neg*-words in Old Russian texts are infrequent but the contexts, where they occur are similar to those in the modern language. (42) is a hybrid MEC sentence conveying the meaning ‘there is no place *x* where *X* can be located’.

³⁸See: <https://www.sport-express.ru/football/rfpl/news/nazarov-napisal-stihotvorenie-posle-razgromaspartaka-ot-zenita-1849910/>.

³⁹There is another approach to the control vs raising distinction, whereby all modal verbs are considered raising predicates that take their subjects from embedded clauses. It is less convenient for our purpose since it does not capture the restructuring effects and the contrast between alethic and deontic modals.

(42) Old Russian

I re^é l'st' est'. **byti**_{INF} **tu** **Izjaslavu**_{DAT} **ně²** **gdě**_{PRED} **polky svoimi v gorax**". i za tēm' ves' polk". Izjaslavl' icēli (Kyeв chronicle [Hypat.], [1150]).

'And he said: "It's a trick: **there is no place for Izjaslav and his troops in the mountains**", and therefore the whole Izjaslav's army escaped without damage.'

The dative NP *Izjaslavu* comes from the infinitival clause *byti svoimi polky v gorax* "to be with one's own troops in the mountains" so that (42) is a raising structure like (17), (18), (27a), (38a) and (39a). The same structure could be construed with an inanimate noun: *zolutu*_{DAT} *bytu*_{INF} *tu ně²gdě*_{PRED} *v gorax* "There is no place for **the gold** to be in the mountains here".⁴⁰ I conclude that based on known Old Russian data, NEG-W—INF sentences originated as embedded DIS structures introduced by impersonal elements with a vacant subject slot filled by the raised dative NP.

4.2 DPS predicatives, *neg*-words and first-person contexts

The status of DPS predicatives as lexical units oriented towards the first person can be checked by the metric introduced in (Zimmerling, 2020a). If most Russian DPS predicatives are *autoreferential*, i.e., oriented towards contacts where the speaker describes his own internal state like 'I am cold', 'I am sad', etc.—it is possible to measure their frequency in the contact position $\langle -1:1 \rangle$ of the subject dative pronoun *mne*_{DAT.ISG} and the predicative. The number of DPS clauses where the pronoun *mne* and the adjacent element assume a subject-predicative relation is called *m*-metrics (*m*). This metrics is used for multiple tasks: 1) ranging a DPS vocabulary over a text collection (Ivanova & Zimmerling, 2019; Zimmerling, 2021a, pp. 253–259); 2) calculating the lower threshold for regular DPS items; 3) checking the status of predicatives by comparing the figures for different person forms. The zero hypothesis for Russian DPS predicatives is that the *m* value is higher than the figure for third-person singular forms *emu*_{DAT.SG.M} & *ej*_{DAT.SG.F} (*e*-metrics, *e*) in the same context $\langle -1:1 \rangle$.

(viii) If *x* is a DPS predicative, $m_x > e_x$, for the same collection of corpus data.

This hypothesis holds true for most Russian DPS predicatives, except for a few items, the lexical meaning of which includes components like 'reproach', cf. *tak i nado* 'X deserved what he got', 'That served X right' (Zimmerling, 2021a, p. 257).⁴¹ The *m*-metric is data-sensitive, the absolute values of *m* result from two factors: 1) the overall frequency of the word in the corpus; 2) the proportion of DPS uses. High values of m_x confirm that *x* belongs to the core DPS vocabulary, while low values of m_x can be due to the undersized corpus and diachronic factors distorting the even distribution of data. E.g., $\langle X-u \rangle$ *nadobno* 'needed' is a bookish word with a high value $m_{nadobno} = 553$ for the whole RNC corpus but is marginal in the XXI century ($m = 9$).

⁴⁰It is possible to analyze (42) as a control structure with *Izjaslavu* generated in the matrix clause. However, one has to take into account the difference in semantics between Old Russian matrix clause dative constructions and presumably raising constructions with a dative argument. The first group includes DIS sentences with an obligatory animate subject and a modal meaning—usually alethic (im)possibility, but also wish, prescription, command or prohibition, cf. *Grigoriju*_{DAT.3G} *goroditi*_{INF} *peregoroda*_{NOM.F} *zaovinnaja*_{NOM.F} 'Gregory has to raise the fence behind the barn'. They do not express statements. The second group includes embedded infinitive clauses with an external head, the dative argument does not discriminate animacy. Here, *Izjaslav* is animate, but the same structure can be built with an inanimate noun *zolutu*. (42) is a statement: the speaker (Prince Vladimir) assumes that his enemy Izjaslav is not in the mountains and asserts that there is no place for him to be there.

⁴¹It is expected that situations where *X* reproaches oneself are less usual than situations where *X* reproaches *Y*.

Table 5 *Neg*-words and DPS predicatives with overt 1st person singular subjects in RNC (2000–2021)

Predicative <i>neg</i> -words					DPS predicatives				
Existential quantification & alethic impossibility					Diverse modal meanings				
	ITEM	m	e	m/e		ITEM	m	e	m/e
1.	<i>nečego</i> ¹	186	124	1,5	1.	<i>nado</i>	1402	892	1,57
2.	<i>nekogda</i>	179	72	2,48	2.	<i>nužno</i>	1226	846	1,23
3.	<i>nekuda</i>	66	72	0,92	3.	<i>ne nado</i>	334	154	2,17
4.	<i>nezačem</i>	19	32	0,59	4.	<i>pora</i>	311	101	3,08
5.	<i>nekomu</i>	19	6	3,17	5.	<i>možno</i>	281	79	3,56
6.	<i>nečego</i> ²	16	23	0,7	6.	<i>nel'zja</i>	209	135	1,5
7.	<i>ne s kem</i>	15	24	0,63	7.	<i>neobxodimo</i>	134	154	0,87
8.	<i>nekogo</i>	9	10	0,9	8.	<i>ne nužno</i>	125	82	1,52
9.	<i>negde</i>	7	26	0,27	9.	<i>položeno</i>	74	62 ⁴²	1,19
10.	<i>neotkuda</i>	3	11	0,27	10.	<i>nedosug</i>	10	7	1,43

To check, whether Russian *neg*-words with higher + ANIM dative subjects are reanalyzed as DPS predicatives, I selected 10 *neg*-words with the highest *m*-values in the sub-corpus of 2000–2021 RNC texts,⁴² calculated their *m/e* ratio and compared them to 10 modal DPS predicatives. Note that despite modal DPS predicatives (CLASS 2 items according to Zimmerling 2021a, p. 248) are similar to *neg*-words they are not synonymic. The *neg*-words express alethic impossibility combined with existential quantification, while DPS items express different kinds of modality without existential quantification. For *nečego*, I calculated the figures for the hybrid *nečego*¹, cf. *nečego*¹ *pit* ‘nothing to drink’ and degraded *nečego*², cf. *nečego*² *spat* ‘one should not sleep’ separately. In the control group of DPS predicatives, 5 items from 10 (*ne nado*, *nel'zja*, *ne nužno*, *neobxodimo* and *nedosug*) contain a lexical negation *ne*, which is not stressed.⁴³

Table 5 shows that the six most frequent DPS items (*m* > 200) and two most frequent *neg*-words (*m* > 100), *nečego*¹ and *nekogda* confirm the hypothesis (viii). Otherwise, the sets of *neg*-words and DPS items are different: 7 *neg*-words from 10 have the *m/e* value < 1, while there is one such item in the DPS set, *neobxodimo*. A plausible explanation is that first-person contexts exclude or reduce the possibility of a raising reading: *mne*_{DAT1.SG} *nado*_{PRED} *vstat*'_{INF} *v vorota*, *mne*_{DAT1.SG} *lučše*_{PRED} *brosit*'_{INF} *pit*'_{INF} can be interpreted only as ‘*X* believes that *X* must stand in the goal’, ‘*X* believes that *X* better stops drinking’, while third person sentences *emu*_{DAT1.3SG.M} *nado*_{PRED} *vstat*'_{INF} *v vorota*, *emu*_{DAT1.3SG.M} *lučše*_{PRED} *brosit*'_{INF} *pit*'_{INF} can also be interpreted ‘*Y* believes that *X* must stand in the goal’, ‘*Y* believes that *X* better stops drinking’.

(43) $\emptyset_Y$ *Nado/lučše* (DAT_X, INF)

The interpretation (43) is not forced by third-person contexts. Most DPS items, cf. *protivno* ‘disgusting’, *žal* ‘sorry’ exclude raising readings, but even those elements that occasionally license embedded DIS sentences like (38) and (39) match (viii) and are oriented towards autoreferential contexts, where *X* is the source of information about *X*’s internal state. E.g., *nado* ‘need’, the most frequent of all DPS items, has the *m/e* value = 1,57. The meanings expressed by the *neg*-words with + ANIM subjects are less transparent and fluctuate between existential quantification and SLP predication. High *m/e* values for *nečego*¹ (1,5), *nekogda* (2,48) and *nekomu* (3,17) prompt that they pattern with DPS predicatives. For *X-u nekogda* ‘*X*

⁴² 43 928 documents, 98 023 229 words (24.08.2022).

⁴³ In standard Modern Russian, *možno* (◇ *p*) and *nel'zja* (∼ ◇ *p*) make up a pair. The positive form **l'zja* ‘allowed’ is lost, while the negative combination *ne možno* ‘not allowed’ is non-standard and survives only as a quotation from XIX century Russian (41 hits in RNS).

has no time', '*X* is busy', the results confirm Apresjan & Iomdin's claim (2010, pp. 102–104) that this item is close to the class of predicatives. For *nečego*¹ and *nekomu* (3,17) an additional factor facilitating a matrix clause reading '*X* believes that *X* is experiencing *p*' is that they take transitive verbs with animate subjects. Five *neg*-words with $m/e < 1$ are either non-autoreferential or not fully integrated into the DPS construction. For the modal items *nečego*² 'should not' and *nezačem* 'no need', 'not for any purpose' the first explanation is likely. To sum up, Russian *neg*-words with + ANIM subjects are reanalyzed as syntactic heads excluding raising readings, but most of them are less integrated into the DPS construction than non-existential modal words like *nado*.

4.3 Summary

Sentences with predicative *neg*-words like (43) originated in Old Russian as embedded DIS clauses but drifted towards finite DPS structures. Embedded DIS sentences with non-identical subjects are ruled out in Modern Russian, while embedded DIS structures with an external impersonal head are possible, cf. raised inanimate subjects *borode negde pomestit'sja, vode nekuda teč', pirogu nado ostyt'*. The animacy restriction in the DPS construction leads to a reanalysis of *neg*-words as morphological and lexical units selecting matrix clause + ANIM subjects. Most DPS items are oriented towards first-person contexts, where raising interpretation is unlikely and show an m_x/e_x value > 1 on text corpora. Some high frequent *neg*-words have similar profiles, but the majority of them occur in third-person contexts. The m/e metric confirms that the reanalysis of *neg*-words as DPS predicatives is not complete.

5 Conclusions

Russian *neg*-words show grammar at its joints: they have different syntax at the start and the end of derivation. Therefore, their descriptions as morphological units and as syntactic agglomerates projecting a bi-clausal structure are not complementary. The surface position of the negation, *k*-word and visualization of overt dative subjects have an impact on semantic interpretation: sentences with *neg*-words do not always have variants with distant negation and do not always reconstruct affirmative correlates. *K*-words raise and make up a new syntactic head with the matrix clause negation, while the embedded infinitival clause is underdeveloped. This leads to clause restructuring and reanalysis of the *neg*-word as a dative predicate. The language-specific trigger of this mechanism is the contrast between finite dative-predicative and non-finite dative-infinitive structures: the dative subjects of predicates are matrix clause elements, while the subjects of dative-infinitive structures are raised. Since embedded dative-infinitive structures have a limited circulation in Modern Russian, the animate subjects of *neg*-words are reanalyzed as matrix clause elements. The Russian dative-infinitive construction does not constrain animacy, the *neg*-words with inanimate subjects, cf. *vode nekuda teč', borode negde pomestit'sja* are remnants of the dative-infinitive syntax.

List of abbreviations

Abbreviations

ACC	Accusative
ADJ	Adjective
AUX	Auxiliary
CP	Complement Phrase

DAT	Dative
DIS	Dative-infinitive-structure
DPS	Dative-predicative-structure
FR	Free Relative
FUT	Future
GEN	Genitive
IMPERF	Imperfective
INDEF	Indefinite
INF	Infinitive
InfP	Infinitive Phrase
INSTR	Instrumental
K-W	<i>k</i> -word (the merged interrogative component of the predicative <i>neg</i> -word)
LOC	Locative
MEC	Modal-existential construction
NEG	Negation
NEG-W	<i>neg</i> -word (a predicative complex with a <i>k</i> -word merged with the general negation)
NEG-WP	Neg-W Phrase (a phrase including a single <i>neg</i> -word)
NiP	<i>Ni</i> -Phrase (a phrase including a single NPI item with the prefix <i>ni</i> -)
NOM	Nominative
NP	Noun Phrase
NPI	negative polarity item
OPT	Optative
PERF	Perfective
PP	Preposition Phrase
PRED	Predicative
PREP	Preposition
PRES	Present
PST	Past
Q	Question
REC	Reciprocal
REFL	Reflexive
TP	Tense Phrase
VP	Verb Phrase
WH	<i>wh</i> -element (different from a <i>k</i> -word)

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Acknowledgements I am grateful to the anonymous reviewers for the valuable critical comments and discussion. The sole responsibility for all shortcomings is mine.

Declarations

Competing Interests The author states that there is no conflict of interest.

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